# Oligopolies in Trade and Transportation: Implications for the Gains from Trade

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**Abstract**

We study how the interplay between oligopoly in the transportation industry and oligopsony power retained by non-atomistic importers affects the transmission of trade policy. Using Chilean customs data, we document strong concentration among carriers and importers and show that freight prices are determined through bilateral bargaining under two-sided market power. We estimate a trade model that endogenizes transportation costs by embedding oligopoly and oligopsony in the transport sector, along with bilateral bargaining. We find sizable carrier markups, partially offset by importer bargaining power. Embedding this mechanism into a general equilibrium trade model, we find that the endogenous response of transportation costs reduces the welfare cost of a 10% tariff by 50% compared to the standard case of iceberg trade costs. This effect is primarily driven by decreasing returns to scale in carriers' supply. Bargaining, in turn, plays a central role in shaping price levels and market allocations in the transportation sector.

**Keywords:** Transport Costs, Bilateral Bargaining, Market Power, Gains From Trade.

**JEL Classification:** F10, F12, F14, R4, D43

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# 1 Introduction

Every year, the transportation sector carries over $20 trillion worth of internationally traded goods around the globe (World Trade Organization, 2024). Recent events, such as the Suez Canal obstruction in 2021, severe port congestion in 2022, piracy attacks in the Red Sea in 2023, and the recent blockage of the Hormuz Strait, have highlighted the indispensable role of the transportation sector in global trade. Yet, relatively little is known about the market structure of this sector and how prices for transportation services are determined. In particular, growing evidence suggests that the transportation market is characterized by large firms providing transport services (Hummels et al., 2009; Ignatenko, 2020; Asturias, 2020; Ardelean and Lugovskyy, 2023), which interact with equally large domestic firms (Bernard et al., 2007; Ciliberto and Jäkel, 2021; Alfaro-Urena et al., 2022).

In this paper, we study transportation prices as the outcome of bilateral negotiations between non-atomistic importers and large carriers. To do so, we provide reduced form evidence that the unit freight prices paid by importers to carriers can be rationalized as the outcome of a bargaining process between two sides of the market, both having substantial market power. We estimate a model of imperfect competition a là Alviarez et al. (2026), adapted to the transportation sector, finding that carriers charge substantial markups to importers. However, importers hold three times more the bargaining power of carriers in determining freight prices, limiting the extent to which carriers can exploit their market power. We then embed these features into a general equilibrium trade model of importing to quantify the welfare effects of import tariffs and transportation cost shocks on consumer prices in the presence of bilateral negotiations in the transportation sector. We show that the presence of endogenous transportation costs reduces the welfare costs of higher tariffs.

We use Chilean import customs-level data spanning from 2007 to 2022 to document new empirical facts about the market structure of the transportation sector and the pricing of international freight services. A key novelty of the dataset is that it records, for each shipment, the carrier responsible for the final leg before customs clearance. This information, combined with freight costs and the mode of transportation, allows us to measure unit freight prices at the shipment level and construct a panel at the importer–carrier level.

We document four key empirical facts. First, we find evidence that both sides of the market, importers and carriers, operate in highly concentrated environments. The top 10% of Chilean importers account for, on average, 95% of total import value. Similarly, the average carrier Herfindahl-Hirschman Index (HHI), computed as the share of imports handled by each carrier within a given origin-HS6-mode triplet, is 0.45, far above the threshold value of 0.25 often used to define an industry as highly concentrated. We interpret this as initial evidence that both carriers and importers possess market power. Second, we find a mean coefficient of variation in unit freight costs within carrier-market-time of approximately 0.9. This implies that a given carrier charges systematically different prices to different importers, consistent with the existence of price discrimination.

Third, we conduct a variance decomposition of unit freight prices using the framework originally developed in the labor literature by Abowd et al. (1999). We find that nearly 90% of the variation in unit freight prices within carrier-market-time is explained by a carrier–importer match-specific component. Lastly, we present reduced-form evidence consistent with both sides exerting market power in the price negotiation of transportation services. Within a given carrier–market pair, unit freight prices decline with the importer's share of the carrier's total shipments and rise with the carrier's share of the importer's total imports. In our baseline specification, we calculate unit freight costs using total weight transported, but we show that our results are not driven by the unit of measurement or by unobserved product characteristics.

These empirical findings suggest that transportation prices are consistent with the outcome of a bilateral bargaining process between importers and carriers, both of whom possess market power.

Next, we borrow the bilateral bargaining framework of Alviarez et al. (2026) and adapt it to the transportation sector, where both carriers and importers hold market power. Carriers' market power arises from their non-atomistic nature and the imperfect substitutability of the services they offer, leading to oligopolistic competition. In contrast, non-atomistic importers exert buyer market power, reflecting the presence of upward-sloping supply curves on the carrier side due to capacity constraints. Crucially, equilibrium unit freight prices are determined via a Nash-in-Nash bargaining. These equilibrium prices are shaped by both the relative bargaining power of the two parties and the strategic incentives originating from oligopolistic and oligopsonistic behavior.

We bring the model to the data to estimate three key parameters that determine the equilibrium transportation prices: the bargaining power of carriers and importers in the negotiation process over the final price, the market power of importers measured through the carriers' scale elasticity, and the market power of carriers proxied by their substitutability. We exploit the network structure of our data, which characterizes importer–carrier relationships, to create valid and relevant IVs and estimate these parameters using a GMM approach.

We estimate within-market substitutability across carriers to be approximately 3.5. This suggests the presence of sizable markups that carriers charge to importers in the transportation sector. Next, we estimate the carriers' supply elasticity to be approximately 0.96, supporting the existence of upward-sloping supply curves for carriers. Additionally, we find that importers have roughly three times more bargaining power than carriers when negotiating the final price. Finally, the combination of an importer markdown of 0.94 and a carrier markup of 2.1 results in a median bilateral markup of 9% over the final transportation price.

Next, we embed the bilateral bargaining framework into a general equilibrium trade model of importing to assess the implications of imperfect competition and bilateral negotiations in the transportation sector for aggregate welfare. The economy consists of a finite number of heterogeneous domestic firms that produce differentiated goods for final consumers. Additionally, the economy features a roundabout production structure, as in Caliendo and Parro (2015). Firms can choose to import a bundle of foreign intermediate inputs, subject to fixed import costs. These imported inputs enhance firm productivity by imperfectly substituting for domestic inputs (Halpern et al., 2015; Blaum et al., 2018). Importing firms negotiate unit freight prices with transportation carriers through Nash-in-Nash bargaining in markets characterized by a finite number of heterogeneous carriers operating under upward-sloping supply curves.

We estimate the model using a combination of customs data and firm-level balance sheet information for the Chilean manufacturing sector. Parameters related to the domestic production process are calibrated using data from the Survey of Manufacturing Industries (ENIA). The remaining parameters are estimated using a two-stage Simulated Method of Moments (SMM), targeting moments from both the domestic economy and the transportation sector.

Lastly, we use the estimated model to quantify the importance of dual market power and bilateral bargaining in the transportation sector for the aggregate economy. We show that the increase in consumer prices due to a 10% tariff shock is approximately 52% higher under standard iceberg trade costs than when transportation costs are endogenous. The attenuation reflects a decline in transportation prices that partially offsets the rise in the factory-gate price of imported goods: higher tariffs reduce demand for imports and thus for transportation services, which in turn lowers transportation prices through decreasing returns to scale. Bilateral markup adjustments play a comparatively minor role. Bargaining power shapes price levels and market allocations in the transportation sector, with its welfare effects operating primarily through equilibrium expenditure shares rather than through the pass-through mechanism directly. Finally, we show that symmetric cost shocks to the transportation sector, such as oil price increases and carbon taxes, are passed through incompletely to consumer prices, with negligible aggregate welfare effects, while asymmetric shocks, such as the port fees proposed under the USTR's Section 301 investigation, generate significant redistribution of market shares and profits across carriers, with similarly small effects on consumer prices.

**Related Literature** This paper relates to at least three strands of literature. First, we contribute to the extensive literature in international trade that studies the determinants of transport costs (Anderson and Van Wincoop, 2003; Eaton and Kortum, 2002). We contribute to this body of work by showing that transportation prices are determined through the interaction between carriers and importers, both of which have market power, and that the equilibrium price results from a bargaining process. This builds on the growing body of evidence suggesting that transport costs are determined endogenously in equilibrium, rather than being externally imposed (Heiland et al., 2019; Brancaccio et al., 2020; Ganapati et al., 2021; Wong, 2022; Do et al., 2024; Tolva, 2025). Early work by Hummels et al. (2009) uses aggregate data to examine the role of market power and price discrimination in shipping. More recent research, enabled by transaction-level data, has explored how freight costs are shaped by factors such as firm size (Ignatenko, 2020), the number of competing shipping firms (Asturias, 2020), and information and search frictions (Ardelean and Lugovskyy, 2023).[1]

[1]Ardelean et al. (2022) surveys recent research on maritime shipping, reflecting the growing availability of micro-level data in this area.

Second, this paper contributes to the literature on firm-to-firm market power by examining importer–carrier relationships. Recent empirical advances in this area have been enabled by the increasing availability of domestic and international firm-to-firm transaction data (Alfaro-Urena et al., 2022; Dhyne et al., 2022). While the theoretical framework follows Alviarez et al. (2026), we adapt it to the transportation sector and embed it in a general equilibrium environment that allows for welfare quantification. On the empirical side, the richness of our data allows us to quantify the market power of both sides, assess their relative bargaining strength, and provide benchmark estimates for key parameters in this industry. We estimate the elasticity of substitution across carriers to be around 3, consistent with previous findings obtained from different settings and econometric approaches (Brancaccio et al., 2020; Asturias, 2020; Wong, 2022; Jeon, 2022). We also find evidence of decreasing returns to scale in carriers' supply, in line with Chen (2024a,b) and Otani (2024). Finally, despite the different context studied in Alviarez et al. (2026), we similarly document a strong bargaining position for importers.

Lastly, we contribute to the literature that quantifies the welfare effects of import tariffs (Arkolakis et al., 2012; Caliendo and Parro, 2015; Blaum et al., 2018). Prior work highlights the importance of market structure for the pass-through of tariffs (Alviarez et al., 2026; Amiti et al., 2019b) and other shocks, e.g. exchange rate fluctuations (Atkeson and Burstein, 2008; Amiti et al., 2019a). We contribute to this literature by showing that the negative impact of import tariffs on domestic consumer welfare is attenuated when transportation costs are endogenous, arising from bilateral bargaining between importers and carriers. Specifically, we quantify how importer market power, carrier market power, and relative bargaining power affect the pass-through of tariffs to domestic price indices.

The rest of the paper is structured as follows: Section 2 provides a set of stylized facts on the transportation industry and bilateral bargaining on unit freight prices. Section 3 presents and estimates the structural model of bargaining over unit freight prices. Section 4 describes the quantitative model, its estimation, and the counterfactual exercises. Section 5 concludes. The Appendices contain additional tables and figures, derivations of key theoretical results, and additional data and estimation details.

# 2 Stylized Facts on Dual-Market Power

## 2.1 Data

We use transaction-level import data from Chilean Customs for the period 2007-2022. Each transaction records the importer, product (HS8 code), mode of transport (sea, air, or road), country of origin, and shipment characteristics including weight (measured in kilograms), quantity, CIF and FOB values, freight and insurance costs. Crucially for our purposes, we also observe the name of the shipping company responsible for transporting the goods. We collapse the data to both the yearly and quarterly level by importer, carrier, country of origin, HS6 product, and transport mode, depending on the exercise.

A key data challenge is carrier identification. Company names are not standardized and are frequently misspelled or recorded inconsistently across transactions. We address this using a combination of string matching algorithms and manual inspection; details are provided in Appendix B.

We focus on imports because a large share of Chilean shipments is organized by importers, consistent with previous work (Ardelean and Lugovskyy, 2023; Teshome, 2018). Customs records include information on which party arranges the shipping contract, encoded via the International Chamber of Commerce's International Commerce Terms (Incoterms). Incoterms divide transactions into two broad categories depending on whether the importer or the exporter is responsible for arranging freight. We restrict attention to importer-arranged transactions to ensure that the parties to the freight price negotiation are exclusively the importing firm and the carrier.[2]

**Structure of Chilean Freight Market** Maritime transport is the most used mode for Chilean imports. Figure E.1 in Appendix E shows that more than 50% of transactions are conducted by sea, consistent with aggregate evidence on international trade and shipping (Ardelean et al., 2022). Air transport is the second most common mode, accounting for roughly 40% of transactions, while road transport is seldom used given Chile's geographical distance from its main trading partners. However, maritime transport dominates in terms of both value and weight, accounting for over 95% of total weight and more than 80% of FOB value (in USD) shipped by sea (left panel of Figure E.1 in Appendix E). The gap between maritime's transaction share and its value and weight share reflects the higher frequency of air shipments, which tend to be smaller and more numerous.

Conditioning on country of origin and HS6-product, more than 80% of importers use a single mode, and this share falls below 50% when conditioning on origin alone, as shown in Figure E.2 in Appendix E.[3]

Despite relying on few transport modes, importers interact with multiple carriers. Indeed, Figure E.3 in Appendix E classifies all carrier-importer pairs within an origin-mode cell into four groups: one carrier to one importer, one carrier to multiple importers, multiple carriers to one importer, and multiple carriers to multiple importers. A large share of matches fall into the many-to-many category, accounting for nearly 40% of imports. Most of the remainder are one-to-many relationships, in which a single carrier serves multiple importers.

[2]Appendix C provides a detailed description of Incoterms and summary statistics on trade flows by category. We show throughout the paper that this sample restriction does not affect the key empirical findings, consistent with Ardelean and Lugovskyy (2023), who find that larger firms face lower freight rates regardless of which party arranges delivery (Proposition 1).

[3]Appendix E provides the sectoral and sourcing composition of Chilean aggregate imports (Figure E.4). Chilean firms tend to source from a limited number of countries (Figure E.5, left panel) and trade a limited number of products (Figure E.5, right panel), consistent with broad patterns in the international trade literature.

## 2.2 Stylized Facts on Market Power in Trade and Transportation

We use transaction-level data to show that both importers and freight carriers act in highly concentrated markets, and that freight prices display patterns consistent with bilateral bargaining and two-sided market power. Throughout this section, a market refers to an origin × mode × HS6-product combination, unless stated otherwise.

**Concentration in Trade and Transportation** We show that both the imports market and the transportation industry are characterized by the presence of large firms.

The left panel of Figure 1 shows that only a handful of firms drive aggregate imports, in line with previous literature (Bernard et al., 2007; Mayer and Ottaviano, 2008; Ciliberto and Jäkel, 2021). The red curve plots the cumulative distribution of imports in 2015 after ranking importers from the left to the right, starting with the biggest. The top 0.1%, 1%, and 10% of importers account for 35%, 65%, and 95% of total imports, respectively.

Freight carriers are equally concentrated. The right panel of Figure 1 plots the average HHI across different market definitions over time. We construct the HHI index by computing the share of imports in weight (kgs) that each carrier ships within a given market. The blue line, which aggregates across products within each origin, reveals that concentration consistently hovers just above the HHI = 0.25 threshold throughout the entire sample period, a level that signals moderate-to-high concentration even under this broader market definition. As expected, narrowing the market definition to the origin-mode-HS6-product level (green line) or the origin-HS6-product level (orange line) yields even higher HHI values, well above 0.4.

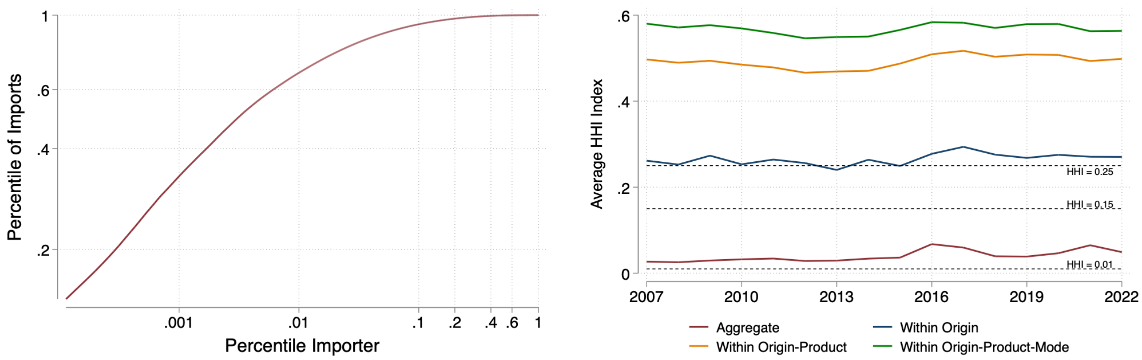
As a robustness check, Figure E.6 in Appendix E shows that concentration among carriers exhibits the same quantitative dynamics if market shares are measured in terms of value (FOB in USD) shipped or number of transactions. In addition, Figure E.7 plots the entire distribution of HHI indices: most of the markets exhibit moderate or high concentration, with indices above the 0.15 and 0.25 thresholds, with no differences between modes of transportation.

**Variation in Bilateral Freight Prices** We document that freight prices vary substantially within market-time cells and within carriers, with the importer-carrier match explaining a large share of this variation. Unit freight prices are constructed at the market-time level by dividing total freight costs by total weight in kilograms. Throughout this section we restrict the sample to importer-arranged transactions identified via Incoterms. Time is measured at an annual frequency.

Unit freight prices are highly dispersed even within carriers, contrary to standard modeling assumptions (Figure 2). For each market-time cell, the mean and median CoV are approximately 0.9 and 0.8, respectively (navy distribution, Figure 2), indicating substantial within-market-time price dispersion. (Ignatenko, 2020; Ignatenko et al., 2025; Ardelean and Lugovskyy, 2023).[4] Crucially, most of this dispersion survives after conditioning on the

[4] Fontaine et al. (2020) and DellaVigna and Gentzkow (2019) define uniform pricing as a CoV below 0.01.

Figure 1: Concentration among Importers and Freight Carriers




**Notes**: The left panel plots the cumulative distribution of importers for the year 2015. Importers are ranked according to their size from left to right on the horizontal axis. The vertical axis reports the cumulative contribution to aggregate imports. Axis are in log scale. The right panel plots the average HHI index across the different markets of the transportation sector over time. Markets are defined according to different levels of granularity. The red line considers a unique aggregate transportation market. The blue line defines markets by the origin country. The orange and green lines define markets as a combination of origin-product and origin-product-mode, respectively. A product is defined as a HS6 category. Modes are sea, air, and road freight. Carriers' market shares are computed in terms of weight (kgs) shipped.

carrier: within market-time-carrier cells, carriers charge systematically different prices to different importers, consistent with the existence of price discrimination.

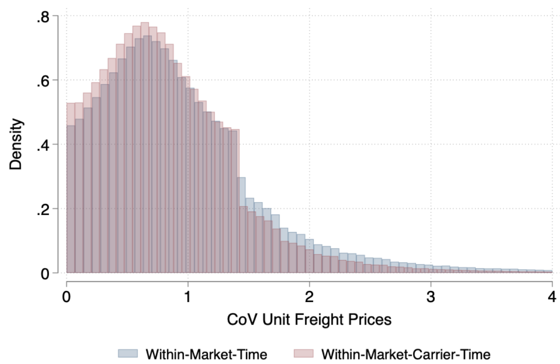
Figure E.8 in Appendix E confirms that these patterns are robust. Price dispersion is quantitatively similar when unit freight prices are measured using shipped value (FOB in USD) or the full transaction sample. The distribution of CoV is also similar across transport modes, with price discrimination equally prevalent in sea and air freight and only slightly lower in road freight. Also, applying the Rauch classification (Rauch, 1999) reveals substantial within-market-time-carrier price dispersion across all product categories, though somewhat weaker for reference-priced goods, ruling out import quality differences as the primary driver of freight price dispersion. We also construct an alternative price measure that accounts for the volumetric structure of freight costs by converting kilograms into Twenty-foot Equivalent Units (TEU), the standard container size in maritime shipping.[5] The price dispersion is quantitatively similar also in this case. Lastly, these patterns are similar when we consider single- or multi-product transactions and when we exclude the smallest and largest five percent of shipments in terms of total FOB.

We follow Fontaine et al. (2020) and consider the following statistical decomposition of

Figure 2 shows that uniform pricing is rare in freight markets.

[5]Since carriers may price shipments based on the volume they occupy rather than their weight alone, the relevant measure of freight capacity depends on the density of the good. In Appendix G, we describe how we construct a product-level (HS6) kg-to-TEU conversion coefficient and use it to express freight costs per unit of container space. We apply this conversion to the subsample of containerized maritime trade, excluding goods that are typically shipped as dry bulk (Arslanalp et al., 2025). Conditioning on maritime trade reduces the sample by roughly half of the transactions but retains around 80% of traded value, as already noted in Figure E.1. Dry bulk is 8% of maritime observations and 13% of its import value. Within this subsample, around 80% of HS6 codes are matched to the TEU conversion mapping.

Figure 2: Unit Freight Price Dispersion




**Notes**: The figure plots the distribution of the CoV of unit freight prices. The navy (maroon) bars correspond to within-market-time (within-market-carrier-time) variation. Markets are defined as a mode-origin-product combination, where modes are sea, air, and road, and products are HS6 categories, respectively. Unit freight prices are computed by dividing total freight cost by the total weight, in kilograms, transported. We restrict our sample to transaction arranged by the importer only. Time is measured at an annual frequency.

unit freight price dispersion to disentangle the contribution of each margin:

$$\log \tau_{ijmt} = FE_i + FE_j + FE_{mt} + \beta \mathbf{X}_{ijmt} + \varepsilon_{ijmt}, \tag{1}$$

where $FE_i$ is an importer fixed effect, $FE_j$ is a carrier fixed effect, $FE_{mt}$ is a market-time fixed effect, and $\mathbf{X}_{ijmt}$ represents a set of control variables such as carriers' experience, age of relationship, and size of transactions. Time $t$ is measured at quarterly frequency.

Firm-level fixed effects alone account for less than 10% of unit freight price variation: the bulk is explained by market-time effects (63%) and the match residual (27%), as shown in Panel A of Table 1. Product and market power heterogeneity across carriers and differences in buyer market power among importers account for a much smaller share of the variance, 4% and 5%, respectively.

Next, Panel B of Table 1 shows that, conditional on carrier and market-time fixed effects, most of the dispersion in unit freight prices is explained by a carrier-importer match-specific component, indicating the presence of bilateral forces in determining freight prices. We find that only 11% of the dispersion can be explained by heterogeneity across importers. The bulk of the variation (89%) is in fact specific to the carrier-importer relationship within a market-year, consistent with the role of bilateral forces in determining bilateral prices, and substantially higher compared to the unconditional variance of 27%.

As a robustness check, Table E.1 in Appendix E shows that the variance decomposition is quantitatively similar when unit freight prices are measured in terms of shipped value, quantity, and for homogeneous goods as defined in Rauch (1999), in the full transaction sample, for single- and multi-product transactions, and when we exclude the smallest and largest five percent of shipments in terms of total FOB. We also construct an alternative price measure that accounts for the volumetric structure of freight costs by converting kilograms into TEU. The variance decomposition is robust to this alternative price definition.

Table 1: Fixed-effect Decomposition of Freight Price Dispersion

| | (1) | (2) |
|---|---|---|
| **Panel A: Share of Price Dispersion Explained by:** | | |
| Observables | . | 0.006 |
| Buyer FE | 0.049 | 0.049 |
| Transport Company FE | 0.033 | 0.033 |
| Product x Time x Origin x Mode | 0.649 | 0.643 |
| Match Residual | 0.270 | 0.269 |
| **Panel B: Within Carrier-Product-Origin-Mode-Time:** | | |
| Observables | . | 0.004 |
| Buyer FE | 0.119 | 0.119 |
| Match Residual | 0.881 | 0.877 |

**Notes**: The table reports the results of a statistical decomposition exercise based on OLS regressions on the estimating specification in Equation (1). Unit freight prices are computed by dividing total freight cost by the kilograms transported. Column (2) includes observable characteristics such as carrier's experience, age of relationship, size of transaction, while Column (1) includes only fixed effects. Markets are defined as a mode-origin-product combination, where modes are sea, air and road, and product are HS6 categories. We restrict the sample to transactions reported as arranged by the importer. Time is measured at quarterly frequency.

**Evidence of Bilateral Bargaining** We provide reduced-form evidence in line with the presence of importer-carrier bilateral bargaining. In the presence of bilateral bargaining, equilibrium prices reflect at the same time both buyer and seller market power (Alviarez et al., 2026; Antràs and Staiger, 2012). We test whether bilateral prices increase in seller market power, proxied by carrier $j$'s share in importer $i$'s total purchases within market $m$ at time $t$, $s_{ijmt}$, and decrease in buyer market power, proxied by importer $i$'s share in carrier $j$'s total shipments within market $m$ at time $t$, $x_{ijmt}$.[6] The mechanism is intuitive: a larger share of importer $i$'s purchases passing through carrier $j$ strengthens the carrier's pricing leverage, while a larger share of carrier $j$'s shipments coming from importer $i$ strengthens the importer's.

We estimate the following empirical specification:

$$\log \tau_{ijmt} = \beta_s \log s_{ijmt} + \beta_x \log x_{ijmt} + \beta \mathbf{X_{ijmt}} + FE + \epsilon_{ijmt}, \tag{2}$$

where $\tau_{ijmt}$ is the unit freight price paid by importer $i$ to carrier $j$ in market $m$ at time $t$, measured as freight costs per kilogram shipped; $\mathbf{X_{ijmt}}$ is a set of control variables including the carrier's experience and the age of the bilateral relationship; and $FE$ represents a set of fixed effects. Time $t$ is measured at quarterly frequency. We construct instruments for bilateral shares to address their endogeneity with respect to bilateral prices, exploiting the network structure of the market (Alviarez et al., 2026). More specifically, we instrument the carrier's seller share $s_{ijmt}$ using the log total sales of carriers other than $j$ to importers other

[6] Formally, the two bilateral shares are defined as:

$$s_{ijmt} = \frac{\text{FOB}_{ijmt}}{\sum_{j'} \text{FOB}_{ij'mt}}, \qquad x_{ijmt} = \frac{w_{ijmt}}{\sum_{i'} w_{i'jmt}},$$

where $\text{FOB}_{ijmt}$ denotes the FOB value of imports shipped by carrier $j$ to importer $i$ in market $m$ at time $t$, and $w_{ijmt}$ denotes the corresponding weight in kilograms.

Table 2: Prices and Bilateral Concentration

| | (1) OLS | (2) OLS | (3) IV |
|---|---|---|---|
| Log Carrier Share | 0.078 | 0.588 | 0.438 |
| | (0.003) | (0.006) | (0.043) |
| Log Importer Share | -0.207 | -0.655 | -0.517 |
| | (0.002) | (0.005) | (0.040) |
| Controls | Yes | Yes | Yes |
| $FE_j + FE_i + FE_{mt}$ | Yes | No | No |
| $FE_{jmt} + FE_{imt}$ | No | Yes | Yes |
| KP F-stat | | | 139.025 |
| $N$ | 1410516 | 950110 | 898329 |

**Notes**: The table reports estimates from the specification in Equation (2). The dependent variable is log unit freight price (freight costs per kilogram). Columns (1) includes carrier, importer, and product-origin-mode-time fixed effects ($FE_j + FE_i + FE_{mt}$). Columns (2) and (3) include carrier-market-time and importer-market-time fixed effects ($FE_{jmt} + FE_{imt}$). Products are HS6 categories and time is measured at quarterly frequency. Column (1) through (3) include controls for the carrier's experience and the age of the bilateral relationship. Columns (1) to (2) report OLS estimates; Column (3) reports IV estimates, where $s_{ijmt}$ and $x_{ijmt}$ are instrumented as described in the text. We exclude all importer-market-time and carrier-market-time singletons from the estimation. Standard errors are clustered at the importer level.

than $i$ in market $m$. For the importer's buyer share $x_{ijmt}$, we use as an instrument the log total purchases of importers other than $i$ from carriers other than $j$ in market $m$.[7]

In line with economic intuition, buyer market power reduces unit freight prices ($\beta_x < 0$), while carrier market power increases unit freight prices ($\beta_s > 0$). In Column (1) of Table 2, which includes importer, carrier, and market-time fixed effects, a one percent increase in the carrier's share is associated with a 0.078 percent increase in unit freight prices, and a one percent increase in the importer's share is associated with a 0.21 percent decrease in unit freight prices. Replacing the three separate sets of fixed effects with importer-market-time and carrier-market-time fixed effects increases the estimated elasticities to 0.59 and 0.66 percent, respectively (Column (2)). Finally, instrumenting for bilateral shares reduces the magnitude of both coefficients relative to their OLS counterparts in Column (3), underscoring the importance of correcting for endogeneity bias.

Table E.2 in Appendix E shows that the qualitative and quantitative results are robust to a range of alternative specifications. We explore differences across transport modes by running the main IV specification separately for sea and air shipments. We also consider several robustness checks with respect to sample and measurement choices: we replace weight-based unit freight prices with quantity-based prices; we restrict the sample to homogeneous goods following the Rauch (1999) classification; we convert quantities to TEU using product-level conversion coefficients; we use the full sample of transactions rather than restricting to those

[7]Formally, the two instruments are defined as the logs of:

$$\widehat{s}_{ijmt} = \sum_{j' \neq j,\, i' \neq i} \text{FOB}_{i'j'mt}, \qquad \widehat{x}_{ijmt} = \sum_{i' \neq i,\, j' \neq j} w_{i'j'mt},$$

where the sums exclude all transactions involving either importer $i$ or carrier $j$, ensuring that the instruments only reflect the trading activity of third parties in the same market.

where shipping is arranged by the importer; and we check that these patterns are similar when we consider single- or multi-product transactions and when we exclude the smallest and largest five percent of shipments in terms of total FOB. Across all specifications, the sign, significance, and broad magnitudes of $\beta_s$ and $\beta_x$ remain stable, supporting the robustness of our findings.[8]

# 3 Estimating Bargaining Power in Transportation Sector

This section derives and estimates a partial equilibrium model of bilateral bargaining in the international shipping market. We focus on the determination of shipping prices through a static Nash-in-Nash bargaining problem between importers and carriers, building on Alviarez et al. (2026). The model allows us to estimate key parameters: the relative bargaining power between importers and carriers, the substitutability across carriers, and the returns to scale of the carriers' production function. The tractability of the framework allows us to embed the key mechanism into a general-equilibrium trade model in Section 4.

While the formal structure of the bilateral pricing equation follows Alviarez et al. (2026), our framework differs along two dimensions. First, the two parties are importers and freight carriers, not importers and exporters of differentiated goods. Accordingly, the upward-sloping supply curve naturally reflects vessel-level capacity constraints in the shipping industry rather than decreasing returns in goods production. Second, as shown in Section 4, we embed the bargaining framework into a general equilibrium environment that links imported inputs and transportation services through a Leontief technology. In this setting, freight prices enter directly into importers' marginal costs and hence affect their gains from trade. The endogeneous transportation cost channel is absent in Alviarez et al. (2026), which abstracts from general equilibrium interactions.

## 3.1 Theory

The market consists of a finite number of importers, indexed by $i$, and a finite number of carriers, indexed by $j$. We denote by $J_i$ the set of carriers serving importer $i$, and by $Z_j$ the set of importers served by carrier $j$. We abstract from endogenous network formation and entry and exit, treating these sets as exogenously given. The model is written for a generic market $m$ at a given time $t$. Market and time indices are suppressed for clarity and reintroduced in the estimation.

**Importers** Each importer $i$ produces one good and sells it domestically facing an isoelastic demand function with elasticity $\sigma > 1$. Importers' output is produced by combining an

[8] In Table E.3, we estimate a variant of Equation (2) that interacts bilateral shares with relationship tenure. The estimated interaction coefficients are statistically insignificant, indicating that the price-share relationships documented in Section 2 are broadly stable across relationships of different tenure.

imported intermediate input, $q_{iF}$, with a domestic input, $q_D$, using a Cobb-Douglas constant-return-to-scale production function with unit substitution elasticity between foreign and domestic inputs. Therefore, the share of imported inputs in total cost and the output elasticity of the imported input are constant, and both are denoted by $\gamma$.[9]

As documented in Section 2, importers organize the shipment and purchase transportation services. We assume that each unit of imported input requires one unit of transportation service to be delivered to the importers. Thus, the imported input $q_{iF}$ used in production can be written as the output of the following Leontief production function:

$$q_{iF} = \min\{\, \overline{q_{iF}}, t_i\}, \tag{3}$$

where $\overline{q_{iF}}$ is the imported input, and $t_i$ is the transportation service purchased by the importer.

We assume that importer $i$'s transportation service, $t_i$, represents a composite bundle of carrier-specific varieties. In other words, each importer $i$ purchases a variety of the transportation service from each carrier $j \in J_i$, combining them with a CES technology. [10] We write:

$$t_i = \left(\sum_{j\in J_i} t_{ij}^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}} \qquad \text{and } \tau_i = \left(\sum_{j\in J_i} \tau_{ij}^{1-\rho}\right)^{\frac{1}{1-\rho}}, \tag{4}$$

where $t_{ij}$ is the quantity of transportation services that importer $i$ purchases from carrier $j$, $\tau_{ij}$ the corresponding bilateral price, and $\rho > 1$ the substitutability across carriers.

It follows that the unit price of imported inputs is $p_{iF} = \overline{p_{iF}} + \tau_i$, where $\overline{p_{iF}}$ is the factory-gate price and $\tau_i$ the price index of the bundle of transportation services. We abstract from any bilateral bargaining between importer and exporter, assuming that the importer is a price taker in the imported input market, taking as given the factory-gate price $\overline{p_{iF}}$.

**Carriers** On the carrier side, we follow Alviarez et al. (2026) and define the production technology in a parsimonious way. Each carrier sells a unique variety of transportation services to all importers in $Z_j$. We assume that the total costs of production are a function of the total output produced by the carrier, denoted by $t_j$: $TC(t_j) = \frac{1}{\zeta_j} t_j^{\frac{1}{\theta}}$, where $\zeta_j$ is a constant capturing productivity differences across carriers, $\theta \in (0,1)$ controls the returns to scale of carriers' production, and $t_j = \sum_{i\in Z_j} t_{ij}$ is the total quantity of transportation services supplied by carrier $j$ across all importers it serves.

Importantly, carriers exhibit an upward-sloped supply curve with marginal cost $c_j$ increasing in quantity, i.e. $\frac{\partial \log c_j}{\partial \log t_j} = \frac{1-\theta}{\theta} > 0$. The presence of decreasing returns to scale in carriers' production guarantees the existence of importers' market power given that the inverse carrier supply elasticity is positive. An upward sloped supply curve represents a reasonable

[9] In other words, $\frac{\partial \log u_i}{\partial \log p_{iF}} = \frac{q_{iF} p_{iF}}{p_i q_i} = \gamma$, where $u_i$ is the marginal cost of importer $i$, while $p_i$ and $q_i$ are the price and the output of importer $i$, respectively.

[10] The CES composite bundle of transportation services can be microfounded *via* a discrete choice model in which the importer chooses, more realistically, one single carrier subject to idiosyncratic taste shock distributed according to a Gumbel distribution, as Anderson et al. (1987) show. This framework delivers the same implications as our composite bundle of carrier-specific varieties assumption.

assumption for the international shipping market: in the presence of capacity constraints, the marginal cost of accommodating an additional shipment rises as vessel-level capacity utilization nears 100 per cent as fully loaded vessels require longer loading and unloading times, ultimately increasing handling costs (Chen, 2024b,a). This assumption is further supported by Dunn and Leibovici (2023), which documents that vessel utilization rates have consistently remained above 90 percent over the past decade.

**Bargaining over shipment prices** We assume that bilateral freight prices are determined via a static Nash-in-Nash bargaining process (Collard-Wexler et al., 2019; Alviarez et al., 2026). The bilateral price $\tau_{ij}$ is the outcome of the following maximization, taking as given all agreements by other importer-carrier pairs:

$$\max_{\tau_{ij}} (\underbrace{\pi_i(\tau_{ij}) - \pi_{i(-j)}}_{\Omega_{ij}})^{\phi} (\underbrace{\pi_j(\tau_{ij}) - \pi_{j(-i)}}_{\Psi_{ij}})^{1-\phi}, \tag{5}$$

where $\phi \in (0, 1)$ is the bargaining weight of the importer, $\Omega_{ij}$ are the gains from trade of importer $i$, the savings from lower per-unit transportation costs net of the cost of purchasing services from carrier $j$, and $\Psi_{ij}$ are the gains from trade of carrier $j$, the extra revenues from serving importer $i$ net of additional production costs. Appendix A.1 provides analytical expressions for $\Omega_{ij}$ and $\Psi_{ij}$ and details of the derivations.

The first-order condition of (5) yields the following closed-form expression for the optimal bilateral price:

$$\tau_{ij} = c_j \mu_{ij} = c_j \left( \omega_{ij} \widehat{\mu}_{ij} + (1 - \omega_{ij}) \overline{\overline{\mu_{ij}}} \right), \tag{6}$$

where $\overline{\overline{\mu_{ij}}}$ is the oligopoly markup, $\widehat{\mu}_{ij}$ is the oligopsony markdown, and $\omega_{ij} \in (0, 1)$ is the effective bargaining weight of the importer. The optimal bilateral markup $\mu_{ij}$ is thus a weighted average of the two polar cases in which one side of the market holds all bargaining power.

$\overline{\overline{\mu_{ij}}} = \frac{\epsilon_{ij}}{\epsilon_{ij}-1}$ is the standard oligopoly markup, where $\epsilon_{ij}$ is the perceived demand elasticity of carrier $j$ with respect to its own price. $\epsilon_{ij}$ depends inversely on $s_{ij} = \frac{\tau_{ij} t_{ij}}{\sum_{z \in J_i} \tau_{iz} t_{iz}}$, the share of carrier $j$ in importer $i$'s total transportation costs: carriers that account for a larger share of an importer's costs face a less elastic residual demand and therefore charge higher markups.[11]

$\widehat{\mu}_{ij} = \theta \frac{1-(1-x_{ij})^{1/\theta}}{x_{ij}}$ is the oligopsony markdown, where $x_{ij} = \frac{t_{ij}}{\sum_{z \in Z_j} t_{zj}}$ is the share of carrier $j$'s total shipments accounted for by importer $i$. For $\theta \in (0, 1)$, $\widehat{\mu}_{ij}$ is strictly decreasing in $x_{ij}$. Importers who account for a larger share of a carrier's business negotiate a lower markdown. $\widehat{\mu}_{ij}$ ranges from $1$ as $x_{ij} \to 0$, when the importer is negligible to the carrier and monopsony power vanishes, to $\theta < 1$ as $x_{ij} \to 1$, reflecting maximum monopsony power.

The effective bargaining weight $\omega_{ij} = \frac{\bar{\phi}\lambda_{ij}}{1+\bar{\phi}\lambda_{ij}}$ captures how bargaining power is shared in equilibrium, where $\bar{\phi} = \phi/(1-\phi)$ is the relative bargaining weight and $\lambda_{ij} = \frac{\sigma-1}{\epsilon_{ij}-1} \frac{\tau_{ij} t_{ij} (\mu_i - 1)}{\Omega_{ij}}$ links the importer's gains from trade to their effective power. Intuitively, $\omega_{ij}$ is increasing in $\bar{\phi}$

[11] Formally, $\epsilon_{ij} = (1 - s_{ij})\rho + s_{ij} \cdot s_{i\tau}(1 - \gamma + \sigma\gamma)$, where $s_{i\tau} = \tau_i / p_{iF}$ is the share of transportation costs in the imported inputs price. This inverse relationship holds whenever $\rho > s_{i\tau}(1 - \gamma(1 - \sigma))$, a condition satisfied for all economically plausible parameter values and verified at our estimates.

and decreasing in $\Omega_{ij}$: when the importer's gains from trade are large relative to the carrier's, the bilateral price is closer to the oligopolistic benchmark $\overline{\overline{\mu_{ij}}}$.

**Discussion** Our framework rests on two important simplifying assumptions. First, our Nash-in-Nash bargaining framework assumes that bilateral prices are determined through a static negotiation process and therefore abstracts from dynamic considerations such as long-term contracting and capacity commitments. Available evidence on contract length in the shipping industry is mixed, with Ardelean and Lugovskyy (2023) documenting that freight rates are typically governed by contracts negotiated annually and Ignatenko et al. (2025) reporting an average contract duration of approximately 45 days in their sample. Our empirical application is at annual frequency, which reduces concerns that short-term contractual dynamics may affect the estimates, although persistent contracts may still matter at this frequency. The estimates should therefore be interpreted as a reduced-form measure of the effective bargaining positions of importers and carriers.

Second, our framework abstracts from the distinction between direct shippers and freight forwarders, who in practice aggregate demand across multiple importers and negotiate rates with carriers on their behalf. Our estimates of importer bargaining power are best understood as reflecting the effective bargaining position of importers in the market, whether exercised directly or through an intermediary.[12]

## 3.2 Estimation

The goal of this section is to estimate the key structural parameters: $\phi$, the relative bargaining power of importers; $\rho$, the substitutability across carriers; and $\theta$, the returns to scale of carriers' production technology. We proceed in two steps. First, we estimate $\rho$ using an IV strategy based on the log-log relationship between prices and shares implied by the model. Second, given the estimated $\rho$, we recover $\phi$ and $\theta$ using the identification strategy in Alviarez et al. (2026). Throughout, we calibrate $\sigma = 6$ and $\gamma = 0.5$ using firm-level data from Chilean manufacturing sectors, as described in Section 4. For claritiy, we now reintroduce market $m$ and time $t$ subscripts suppressed in Section 3.1.

**Identification of $\rho$** The substitutability across carriers, $\rho$, is identified from the demand equation for transportation services. Equation (4) implies that, for importer $i$ in market $m$ at time $t$, the log share of carrier $j$ in total transportation costs satisfies:

$$\log s_{ijmt} = -(\rho - 1)\left(\log \tau_{ijmt} - \log \tau_{imt}\right) + \nu_{ijmt}, \tag{7}$$

[12]Recent evidence in Ignatenko et al. (2025) using data on a single route shows that this distinction may be quantitatively important: following the COVID-19 demand surge, freight forwarders faced significantly larger increases in freight rates than direct shippers, reflecting differential exposure to carrier market power. Unfortunately, data limitations prevent us from distinguishing between transactions arranged by direct shippers and those intermediated by freight forwarders within our sample, and we leave a fuller treatment of this distinction as an important avenue for future research.

where $\tau_{imt}$ is the importer-level price index and $\nu_{ijmt}$ is an idiosyncratic demand shock, assumed *i.i.d.* across $i$, $j$, $m$, $t$ with conditional mean zero. Assuming $\rho$ is constant across markets and importers, this translates into the estimating equation:

$$\log s_{ijmt} = \beta \log \tau_{ijmt} + \alpha_{imt} + \nu_{ijmt}, \tag{8}$$

where $\alpha_{imt}$ denotes importer-market-time fixed effects and $\beta = -(\rho - 1)$ is the coefficient of interest.

We estimate Equation (8) by differencing out the importer price index $\tau_{imt}$, which is common across all carriers serving importer $i$ in market $m$ at time $t$ (Broda and Weinstein, 2006; Feenstra, 1994). Defining $\Delta \log v_{ijj'mt} \equiv \log v_{ijmt} - \log v_{ij'mt}$ for any variable $v$, Equation (7) becomes:

$$\Delta \log s_{ijj'mt} = -(\rho - 1)\Delta \log \tau_{ijj'mt} + \Delta\nu_{ijj'mt},$$

which absorbs importer-market-time fixed effects by construction. For each importer, we use the carrier with the median buyer share as the reference carrier $j'$.

To address the endogeneity of bilateral prices, we employ two sets of instruments. First, we use Hausman-type instruments: the log price charged by the same carrier $j$ to other importers in other markets (Hausman et al., 1994). These exploit common carrier-level cost shocks across markets, under the assumption that demand shocks are uncorrelated across markets, cov$(\nu_{ijmt}, \nu_{i'jm't}) = 0$.[13] Second, we include the log number of carriers and importers active in each market as market structure instruments, exploiting variation in competitive conditions under the assumption that entry decisions are predetermined with respect to demand shocks (Berry et al., 1995; Gandhi and Nevo, 2021). Finally, we include carrier-market and time fixed effects to absorb time-invariant unobserved heterogeneity across carriers, limiting residual endogeneity to time-varying pair-specific shocks.

**Identification of $\phi$ and $\theta$** We follow Alviarez et al. (2026) and Dhyne et al. (2022) to identify the bargaining weight $\phi$ and the carriers' scale parameter $\theta$. From Equation (6), the log bilateral freight price satisfies:

$$\log \tau_{ijmt} = \log \mu_{ijmt} + \log c_{jt} + \nu_{ijmt},$$

where $\nu_{ijmt}$ is mean-zero i.i.d. and captures unobserved cost differences across importers of a given carrier, reflecting, for instance, quality differentiation or shipment customization. Taking the difference between the prices carrier $j$ charges to any two importers $i$ and $k$ cancels

[13] This assumption would be violated by carriers' unobserved promotional campaigns across markets. We do not view this as a relevant concern in international shipping, where pricing is driven by route-specific factors such as distance and port traffic rather than demand-side marketing.

the marginal cost $\log c_{jt}$ and yields the moment condition:

$$g(\phi, \theta, \Lambda_{jikmt}) \equiv \mathbb{E}\left[\log \tau_{ijmt} - \log \mu_{ijmt} - (\log \tau_{kjmt} - \log \mu_{kjmt}) \mid \Lambda_{jikmt}\right] = 0 \quad \forall\, i, k, j, m, t, \tag{9}$$

where $\Lambda_{jikmt}$ contains the bilateral shares $s_{ijmt}$, $s_{kjmt}$, $x_{ijmt}$, $x_{kjmt}$, $s_{i\tau mt}$, $s_{k\tau mt}$ and the corresponding instruments for pairs $i$-$j$ and $k$-$j$. Identification of $\phi$ and $\theta$ is guaranteed by the strict monotonicity and invertibility of $g(\cdot)$ in both parameters, and by the nonlinearity of $\mu_{ijmt}$ in the bilateral shares (Alviarez et al., 2026).

We estimate the parameters by IV-GMM:

$$\min_{\phi,\theta}\; G(\phi,\theta)'\, Z\, W\, Z'\, G(\phi,\theta), \tag{10}$$

where $G(\phi, \theta)$ stacks the moment conditions across all $i$-$k$-$j$-$m$-$t$ cells, $Z$ is the matrix of instruments, and $W$ is the optimal weighting matrix, estimated in a first step under the identity matrix.

Although differencing removes carrier marginal costs, differences in unobserved cost components may still be correlated with bilateral shares, creating endogeneity. We address this using Hausman-type instruments in $Z$: the mean and median buyer and seller shares in the same market, computed excluding the pairs $i$-$j$ and $k$-$j$ involved in each moment (Hausman et al., 1994; Alviarez et al., 2026). We further demean the moment conditions at the market-time-importer level, so that residual endogeneity is limited to time-varying pair-specific shocks.

**Data construction** We estimate the parameters using the full sample from 2007 to 2022. In our baseline specification, we construct the key variables $s_{ijmt}$, $x_{ijmt}$, $s_{i\tau mt}$, and $\tau_{ijmt}$ at the importer-carrier-market-year level, where a market is an HS6-product-origin-mode combination as defined in Section 2.2. Unit freight prices $\tau_{ijmt}$ and carrier shares $x_{ijmt}$ are computed in terms of shipped weight (in kilograms). Importer shares $s_{ijmt}$ are measured as expenditure shares (total freight cost to carrier $j$ divided by total freight cost to all carriers).[14]

In addition to the cleaning described in Section 2.1, we apply the following sample restrictions. We drop observations with zero bilateral shares, unit freight prices, or transport-cost shares. We trim unit freight prices at the 5% level within each market and separately at the 5% level in the full sample. We retain only importer-carrier pairs that trade for at least two consecutive years, to reduce the influence of occasional and lumpy importers. For the estimation of $\phi$ and $\theta$, we exclude carriers transacting with only one importer within a market, since the moment condition in Equation (9) is not defined in that case, and retain only markets with at least three active carriers and importers trading with at least two carriers. For the estimation of $\rho$, we additionally exclude carriers operating in only one market or serv-

[14] We aggregate all transactions at the importer-market-year level and construct the share of transportation costs in the price of imports as $s_{i\tau mt} = \frac{\sum_{j \in J_{imt}} \text{Transportation Cost}_{ijmt}}{\sum_{j \in J_{imt}} (\text{FOB}_{ijmt} + \text{Transportation Cost}_{ijmt})}$, where $J_{imt}$ is the set of carriers serving importer $i$ in market $m$ at time $t$.

Table 3: Summary Statistics

| | Mean | Std. |
|---|---|---|
| Log $\tau_{ijmt}$ | -1.372 | 0.725 |
| Importer's Share $s_{ijmt}$ | 0.425 | 0.226 |
| Carrier's Share $x_{ijmt}$ | 0.425 | 0.313 |
| Transport Share $s_{i\tau mt}$ | 0.099 | 0.088 |
| Number of Carriers per Market | 8.387 | 4.579 |
| Number of Importers per Market | 26.297 | 29.184 |
| Number of Carriers per Importer | 3.014 | 0.957 |
| Number of Importers per Carrier | 7.667 | 1.902 |

**Notes**: The table reports means and standard deviations for key variables across markets, weighted by total weight transported (in kilograms). $\tau_{ijmt}$ is the unit freight price paid by importer $i$ to carrier $j$ in market $m$ at time $t$, computed as total freight cost divided by kilograms transported. $s_{ijmt}$ is the share of carrier $j$ in importer $i$'s total transportation costs in market $m$ at time $t$. $x_{ijmt}$ is the share of importer $i$ in carrier $j$'s total quantity transported in market $m$ at time $t$. $s_{i\tau mt}$ is the share of transportation costs in the price of imports at the importer-market-time level. Markets are as defined in Section 2.2.

ing only one importer, since the Hausman-type instrument requires variation across markets, and retain only importers purchasing from more than one carrier within a market, as the differencing strategy requires at least two carriers per importer. Finally, we drop observations implying a perceived demand elasticity $\epsilon_{ijmt} < 1.1$, which is inconsistent with the model.[15]

Table 3 reports summary statistics for the estimation sample. Bilateral freight prices are highly dispersed and concentration is high in both the import and transportation markets. The average market has 26 importers and 8 carriers, with each importer connected to an average of 3 carriers and each carrier to 8 importers, generating high and dispersed bilateral shares $s_{ijmt}$ and $x_{ijmt}$. The average share of transportation costs in the price of imports, $s_{i\tau mt}$, is 10%, underscoring the quantitative relevance of freight costs for importers.

### 3.3 Results

This section presents the estimation results, their robustness, their heterogeneity across markets, and the implied bilateral markups.

**Main estimates** Table 4 reports the baseline estimates. Column (1) reports the estimated coefficient from Equation (8) and the implied substitutability across carriers, $\rho$. Columns (2) and (3) report the estimated coefficients from the GMM in Equation (10), together with the implied bargaining parameter $\phi$.

Carrier substitutability is low, with $\widehat{\rho} = 3.55$ (Table 4, Column (1)), enabling carriers to charge substantial oligopolistic markups. [16] The estimated value is within the range reported

[15] See Footnote 11.

[16] Table F.1 in Appendix F shows that the OLS estimate of the price elasticity is positive, displaying a bias towards zero due to the positive correlation between demand and price shocks (Column (1)). The strong negative value of the CES elasticity estimated in the main specification supports the validity of our instruments in correcting for this endogeneity bias. In addition, the instruments have no residual predictive power for quantities conditional on prices: a joint F-test of instrument exclusion in the share equation yields a p-value of

in the literature using different data and identification strategies.[17]

The estimated relative bargaining power is $\widehat{\tilde{\phi}} = 3.1$, implying $\widehat{\phi} = 0.75$: on average, importers enjoy substantial bargaining power, allowing them to put relatively more weight on the markdown. The estimated scale elasticity is $\widehat{\theta} = 0.49$, well below one, implying a carriers' supply elasticity $\frac{\theta}{1-\theta} \approx 0.96$ and confirming that importers exert buyer market power on carriers.[18] Both parameters are precisely estimated.

We perform several exercises to assess the robustness of our estimates. Table F.1 in Appendix F shows that the estimated substitutability across carriers is robust to alternative sets of fixed effects. Table F.2 shows that segmenting the instruments does not affect the estimate of $\rho$. Tables F.3 and F.4 report comparable estimates across all three parameters when (i), for containerized maritime trade, prices are defined in terms of TEUs rather than weight, (ii) less restrictive data cleaning is applied, and (iii) markets are defined as origin–mode rather than origin–mode–product (iv) when we restrict the sample to single- and (v) multi-product transactions and (vi) when we exclude the bottom and top five percent of the shipment size distribution measured by FOB value. Finally, we show that the estimate of $\rho$ is robust to the choice of reference carrier in Equation (8) by using the second-lowest-share carrier. Across all alternative specifications, the estimates for $\widehat{\rho}$, $\widehat{\phi}$, and $\widehat{\theta}$ are quantitatively close to our baseline values. A few specifications, namely the Hausman-only and Hausman $+ N_i$ columns in Table F.2 and the alternative-market specification in Table F.3, have weaker first stages, with estimates close to baseline but less precisely identified.

**Heterogeneity across markets** We estimate the full vector of parameters for each individual origin-HS6-product-mode market.[19]

Table 5 reports selected moments of the distribution of estimated parameters across markets. Among these, only the median market estimate of $\widehat{\theta}$ closely matches the baseline value reported in Table 4, whereas the median market estimates of $\widehat{\rho}$ and $\widehat{\phi}$ are somewhat lower than their baseline counterparts. There is also substantial heterogeneity across markets: the interquartile range is 3.5 for $\rho$, and 0.4 for both $\phi$ and $\theta$. Figure F.1 in Appendix F shows that the distributions of all three parameters are quantitatively similar across modes of transportation.

The estimated parameters correlate with observable market characteristics in ways con-

0.6.

[17] Wong (2022) and Jeon (2022) estimate values around 3 using the round-trip effect and ship size and age as instruments, respectively. Brancaccio et al. (2020) and Asturias (2020) estimate elasticities between 5 and 6 for the dry bulk and container sectors, respectively. Chen (2024b) and Chen (2024a) use the Houthi attacks in 2023 as an instrument and estimate a smaller elasticity of around 1.2.

[18] Chen (2024b) and Otani (2024) strongly reject constant and decreasing marginal costs when analyzing the effects of shipping alliances and cartels, respectively. Chen (2024a) estimates a supply elasticity of around 2.2 using CTS data on vessel capacity.

[19] We apply the baseline specification within each market. If $\widehat{\beta}$ is positive or fails to converge, we fall back sequentially to: (i) the Hausman instrument alone, (ii) the number of carriers and importers as the sole instruments, and (iii) estimation in levels with importer-time-market fixed effects using either instrument specification. If the estimation of $\phi$ and $\theta$ fails to converge, we use only the mean buyer and seller bilateral shares as instruments, if that also fails, we use only the median buyer and seller bilateral shares.

Table 4: Estimated Model Parameters

| | $\widehat{\beta}$ | $\widehat{\overline{\phi}}$ | $\widehat{\theta}$ |
|---|---|---|---|
| Estimate | -2.548 | 3.117 | 0.492 |
| | (0.580) | (1.453) | (0.044) |
| Implied $\widehat{\rho}$ | 3.548 | | |
| Implied $\widehat{\phi}$ | | 0.757 | |
| Fixed Effects | $FE_j \times FE_m + FE_t$ | $FE_i \times FE_m \times FE_t$ | $FE_i \times FE_m \times FE_t$ |
| KP $F$-stat | 14.543 | 2104.252 | |
| Hansen $J$-stat | 0.444 | 2.036 | |
| Hansen $J$ $p$-value | 0.801 | 0.361 | |
| $N$ | 145652 | 835045 | 835045 |

**Notes**: Column (1) reports the estimated price elasticity from Equation (8), estimated in differences using as instruments the log price charged by carrier $j$ to other importers in other markets and the log number of carriers and importers competing in each market. Standard errors are clustered at the importer level. Columns (2) and (3) report the estimated relative bargaining power $\overline{\phi}$ and scale elasticity $\theta$ from Equation (9), using the mean and median buyer and seller bilateral shares in the market, excluding the pairs involved in each moment, as instruments. Standard errors are robust. Moment conditions are demeaned at the importer $\times$ market $\times$ year level before estimation. Markets are defined as HS6-product-origin-mode cells. We drop observations with $|\log \tau_{ijmt} - \log \tau_{kjmt}| > 1$. Implied $\phi$ is recovered from $\widehat{\overline{\phi}}$ using $\overline{\phi} = \frac{\phi}{1-\phi}$.

Table 5: Heterogeneity across Markets

| | $\widehat{\rho}$ | $\widehat{\phi}$ | $\widehat{\theta}$ |
|---|---|---|---|
| Mean | 5.06 | 0.58 | 0.45 |
| Weighted Mean | 5.28 | 0.52 | 0.49 |
| Median | 2.60 | 0.61 | 0.47 |
| IQR | 3.50 | 0.46 | 0.39 |
| Number of Markets | 2197 | 1551 | 1551 |

**Notes**: The table reports moments of the distribution of $\widehat{\rho}$, $\widehat{\phi}$, and $\widehat{\theta}$ estimated at the origin-HS6-product-mode level. $\widehat{\rho}$ is estimated from Equation (8); $\widehat{\phi}$ and $\widehat{\theta}$ from Equation (9). Weighted moments use total weight shipped in each origin-HS6-product-mode triplet as weights. Markets with $\widehat{\rho} > 50$ are excluded. The smaller number of markets for $\widehat{\phi}$ and $\widehat{\theta}$ reflects estimation requirements that exclude markets with too few observations.

sistent with economic intuition, supporting the validity of our estimates. As reported in Table F.5 in Appendix F, importer bargaining power $\phi$ tends to be lower in markets with more importers, while its relationship with the number of carriers is imprecisely estimated and close to zero. The scale elasticity $\theta$ moves in the opposite direction, increasing with the number of importers and decreasing with the number of carriers, a pattern consistent with importers exerting greater buyer market power when they face fewer competitors. Carrier market power, which is inversely related to $\rho$, appears lower in markets with more carriers and higher in markets with more importers. We find broadly similar patterns, even if weaker and less precisely estimated, when replacing competitor counts with the HHI indices of the bilateral shares $s_{ijmt}$ and $x_{ijmt}$.

**Implied bilateral markups** We use the estimated parameters from Table 4 to compute implied bilateral markups and their components. Table 6 reports moments from the distribution of market-level average bilateral markups, together with the underlying oligopolistic markup, oligopsonistic markdown, and bargaining weight. The mean and median bilateral markups are 8 and 9 percent, respectively, consistent with estimates for the transportation industry in De Loecker and Eeckhout (2017). The oligopolistic markup factor ranges from 1.8 to 2.2 across markets, reflecting the low estimated demand elasticity, while the oligopsonistic markdown ranges from 0.90 to 0.96. The bilateral markup combines these two forces, with an average bargaining weight of 80 percent on the oligopsonistic markdown. At the pair level, Figure F.2 in Appendix F shows a strong negative correlation between oligopolistic markups and oligopsonistic markdowns across importer-carrier pairs, driven by the positive correlation between buyer and seller bilateral shares $s_{ijmt}$ and $x_{ijmt}$.

Table 6: Bilateral Markups

| | Mean | p25 | p50 | p75 |
|---|---|---|---|---|
| Oligopolistic Markup | 2.143 | 1.861 | 2.020 | 2.260 |
| Oligopsonistic Markdown | 0.921 | 0.894 | 0.935 | 0.963 |
| Bargaining Weight | 0.808 | 0.797 | 0.806 | 0.817 |
| Bilateral Markup | 1.080 | 1.057 | 1.091 | 1.113 |
| $N$ | 29655 | | | |

**Notes**: The table reports moments from the distribution of origin-HS6-product-mode-time average oligopolistic markups, oligopsonistic markdowns, bilateral markups, and bargaining weights. All objects are constructed using the estimated parameters from Table 4 and the estimation sample. Moments are weighted by the number of importer-carrier pairs in each origin-HS6-product-mode cell.

# 4 Aggregate Implications

In this section, we embed the bargaining framework developed in Section 3 into a general-equilibrium model of importing to quantify the aggregate effects of bilateral bargaining in the international shipping market and its role in shock transmission. Appendix A.2 provides additional derivation details.

## 4.1 Theory

**Consumption and Demand** The economy is populated by a unit measure of consumers who supply $L$ units of labor inelastically and consume a bundle $C$ over $N$ differentiated domestic products:

$$C = \left( \sum_{i=1}^{N} c_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \tag{11}$$

where $\sigma > 1$ is the elasticity of substitution across varieties. Consumers maximize utility subject to $\sum_{i=1}^{N} p_i c_i \leq wL + \sum_{i=1}^{N} \pi_i$, where $w$ is the wage and $\pi_i$ are firm profits. This yields demand $c_i = p_i^{-\sigma} P^{\sigma} Y$, where $P = \left(\sum_{i=1}^{N} p_i^{1-\sigma}\right)^{1/(1-\sigma)}$ is the aggregate price index and $Y$ aggregate income.

**Firms and Input Trade** Each product $i \in \{1, \ldots, N\}$ is produced by a single firm under monopolistic competition, combining labor $l_i$ and an intermediate input bundle $x_i$ via a CRS Cobb-Douglas technology:

$$y_i = \varphi_i \, l_i^{1-\beta} \, x_i^{\beta}, \tag{12}$$

where $\varphi_i$ is firm-level productivity and $\beta \in (0, 1)$ is the intermediate input share. The intermediate input bundle $x_i$ combines domestic and foreign intermediates $q_{iD}$ and $q_{iF}$ via a CES aggregator:

$$x_i = \left(\eta \, q_{iD}^{\frac{\xi-1}{\xi}} + (1-\eta) \, q_{iF}^{\frac{\xi-1}{\xi}}\right)^{\frac{\xi}{\xi-1}}, \tag{13}$$

where $\eta \in (0, 1)$ is the CES weight on the domestic intermediate and $\xi > 1$ is the elasticity of substitution between domestic and foreign intermediates.

Importing requires a fixed cost of $f$ units of domestic labor. A firm imports only when the cost reduction from foreign intermediates, via love-of-variety gains, outweighs this fixed cost (Halpern et al., 2015; Gopinath and Neiman, 2014; Antras et al., 2017).

Following Caliendo and Parro (2015), we adopt a roundabout production structure in which the domestic intermediate $q_{iD}$ is itself a CES bundle of all domestic firms' output: $q_{iD} = \left(\sum_{v=1}^{N} y_{iv}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$, where $y_{iv}$ is firm $v$'s output purchased by firm $i$. The domestic input price $p_D$ is therefore endogenous, ensuring that non-importing firms are also affected by changes in the transportation sector through their purchases of inputs from importers.

Building on Section 3, importing firms purchase transportation services according to Equation (3) and a generalization of Equation (4) that includes match-specific taste weights $\alpha_{ij}$ (see Appendix A.2). There exists a unique transportation market, that is a single route from the rest of the world to the domestic economy, populated by a finite number of carriers. The total cost of production for carrier $j$ is $TC(t_j) = \frac{1}{\zeta_j} t_j^{\frac{1}{\theta}}$, where $\zeta_j$ is a carrier-specific productivity shifter and $\theta \in (0, 1)$ governs returns to scale, consistent with the micro-level model in Section 3. Bilateral freight prices are determined via the Nash-in-Nash bargaining process in Equation (5). The bilateral price formula in Equation (6) retains its structure, but the gains-from-trade terms $\Omega_{ij}$ and the weight $\lambda_{ij}$ are recomputed using the CES aggregator in Equation (13) (see Appendix A.2 for the modified expressions).

The firm's profit maximization problem is:

$$\pi_i = \max \left\{ u_i^{1-\sigma} \times B - wf \, \mathbf{1}(q_{iF} > 0) \right\}, \tag{14}$$

where $u_i$ is the unit cost of production and $B \equiv \frac{1}{\sigma} \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} P^{\sigma-1} S$ is a demand shifter common across firms, with $S$ denoting aggregate spending.[20] The unit cost is:

$$u_i = \frac{1}{\varphi_i} w^{1-\beta} p_{x,i}^{\beta} = \frac{1}{\varphi_i} w^{1-\beta} \left( \eta^{\xi} p_D^{1-\xi} + (1-\eta)^{\xi} \left( \overline{p}_{iF} + \tau_i \right)^{1-\xi} \right)^{\frac{\beta}{1-\xi}}, \tag{15}$$

where $p_{x,i}$ is the price index of the intermediate bundle $x_i$, $\overline{p}_{iF}$ is the factory-gate price of the imported input, and $\tau_i = \left( \sum_{j \in J_i} \alpha_{ij} \tau_{ij}^{1-\rho} \right)^{\frac{1}{1-\rho}}$ is the CES price index of transportation services, generalizing Equation (4) with match-specific weights $\alpha_{ij}$.

**General Equilibrium** Equations (11)-(14) above describe firms' optimal decisions. We close the model in general equilibrium, imposing the equilibrium in the labor market and balanced trade between the domestic economy and the rest of the world. Assuming $w = 1$, balanced trade requires that aggregate exports equal total imported intermediate inputs:

$$\sum_i^N p_i y_i^{ROW} = \sum_i^N (1 - s_{iD}) m_i,$$

where $m_i$ denotes total intermediate input spending of firm $i$, and $(1 - s_{iD})$ the share of spending on imported inputs.

An equilibrium is defined as a set of (bilateral) prices $\{w, [p_i], [\tau_{ij}]\}$, labor demands for production and fixed costs, demand for services $[t_{ij}]$, production and consumption $\{[y_i], [c_i], [y_i^{ROW}]\}$, and input demands $\{[q_{iD}], [q_{iF}]\}$ such that firms maximize profits, consumers maximize utility, trade is balanced, and labor and goods markets clear.[21]

## 4.2 Calibration and Estimation

We now parametrize the model using Chilean customs and micro data. Tables 7 and 8 summarize the parameters and the calibrated values or the moments used in the estimation.

### 4.2.1 Externally Calibrated Parameters

**Bargaining and transportation sector.** We use the estimates from Section 3 to parametrize the bargaining process and the transportation sector. We set the elasticity of substitution across carriers to $\rho = 3.5$, the bargaining power to $\phi = 0.75$, and the carriers' return to scale to $\theta = 0.5$. From Table 3, we set the number of carriers in the transportation sector equal to 3, the average number of carriers per importer in the data, and the number of importers equal to 8, the average number of importers per carrier.[22]

[20] Aggregate spending satisfies $S = \frac{\sigma}{(1-\beta)(\sigma-1)} \left( L - \sum_{i=1}^N f\, \mathbf{1}(q_{iF} > 0) \right)$.

[21] Formally, the labor market clearing condition is: $L = \sum_i (l_i + f \mathbf{1}(q_{iF} > 0))$. Similarly, the good market clearing condition for each firm $i$ is: $y_i = c_i + y_i^{ROW} + \sum_v^N y_{iv}$.

[22] We calibrate the model to the average number of carriers per importer and importers per carrier because this is consistent with the CES structure of the model, in which each importer is connected to each carrier. The market-level averages reported in Table 3 reflect an incomplete network, as agents do not necessarily trade with all other agents in the market. Capturing the formation and adjustment of these relationships would require

**Domestic Economy.** We use firm-level balance sheet data from the Survey of Manufacturing Industries (ENIA) from 1995 to 2018 to calibrate the parameters governing the domestic economy and the production process ($\sigma$, $\beta$, and $\xi$). We follow Oberfield and Raval (2021) and identify the elasticity of substitution $\sigma$ from firms' profit margins: $\frac{\text{Revenue}_i}{\text{Cost}_i} = \frac{\sigma}{\sigma-1}$, where costs are computed as the sum of the wage bill, materials and electricity expenditure, and the user cost of capital. We set $\sigma = 6$, close to the median value of 5.9 in the manufacturing sector. Given $\sigma$, the observed material spending share identifies $\beta$ via $\frac{m_i}{p_i y_i} = \beta \frac{\sigma-1}{\sigma}$, where $m_i$ denotes total intermediate input spending of firm $i$. We set $\beta = 0.45$, rounding up the median material spending share of 0.427 in the manufacturing sector. Finally, we identify the substitutability between domestic and imported inputs $\xi$ by noting that firm output can be written as:

$$y_i = A\, \varphi_i\, l_i^{1-\beta}\, m_i^{\beta}\, s_{iD}^{-\frac{\beta}{\xi-1}},$$

where $A$ collects all general equilibrium variables and $s_{iD}$ is the share of spending on domestic inputs. We exploit variation in domestic expenditure shares holding material spending fixed to identify $\xi$ (Blaum et al., 2018; Zhang, 2017), using standard production function estimation techniques (Levinsohn and Petrin, 2003; Ackerberg et al., 2015). This yields an estimate of $\widehat{\xi} = 3.77$, which we round to $\xi = 4$ in the quantitative model. The calibrated values are consistent with previous estimates in the literature (Blaum et al., 2018; Alviarez et al., 2026; Halpern et al., 2015). Appendix D provides additional details on the ENIA dataset, its cleaning, and the calibration procedure.

### 4.2.2 Internally Estimated Parameters

**Number of firms and fixed costs of importing.** The fixed cost of importing, $f$, is pinned down by the share of importing firms in the economy. Chilean manufacturing microdata show that 20% of domestic firms are importers. We set the number of domestic firms to $N = 40$, so that the number of importers in the transportation sector is consistent with this empirical share. The fixed cost $f$ is then estimated to match this importing share in equilibrium.

**Firm and carrier productivity, and carrier-importer matching shocks.** Four parameters govern the distributions of firm and carrier heterogeneity. Carrier productivity $\zeta_j$ is drawn from a log-normal distribution with median normalized to one and variance $\sigma_\zeta^2$. Firm efficiency $\varphi_i$ is drawn from a log-normal distribution with mean $\mu_\varphi$ and variance $\sigma_\varphi^2$. Match-specific taste shocks in the transportation sector, $\alpha_{ij}$, are drawn from a log-normal distribution with unit mean and variance $\sigma_\alpha^2$. We estimate these four parameters, $\sigma_\zeta^2$, $\mu_\varphi$, $\sigma_\varphi^2$, and $\sigma_\alpha^2$, by targeting salient features of the empirical distributions of sales and bilateral shares. The dispersion in domestic sales and carrier size identifies the efficiency variances $\sigma_\varphi^2$ and $\sigma_\zeta^2$, respectively. The mean $\mu_\varphi$ captures the average productivity differential between importers

introducing fixed costs of establishing links or search and switching dynamics, which is beyond the scope of our modeling framework. We view this as a simplifying assumption and leave a richer treatment of network formation to future work.

Table 7: Calibrated Parameters

| **Panel A: Calibrated Parameters** | | |
|---|---|---|
| Transportation Sector and Bargaining Process | | |
| $\rho$ | 3.5 | Estimated from Section 3.3 |
| $\theta$ | 0.5 | Estimated from Section 3.3 |
| $\phi$ | 0.75 | Estimated from Section 3.3 |
| $N_j$ | 3 | Table 3 |
| $N_i$ | 8 | Table 3 |
| Domestic Economy | | |
| $\beta$ | 0.45 | Median Share of Materials |
| $\xi$ | 4 | Estimated using Production Function |
| $\sigma$ | 6 | Median Markup |
| $N$ | 40 | Share of Importers |
| $\eta$ | 0.5 | Normalized |
| L | 1 | Normalized |

**Notes**: The table reports calibrated parameter values. Parameters are calibrated using Chilean customs transaction data from 2007 to 2022 and the Survey of Manufacturing Industries (ENIA) from 1995 to 2018. Additional details on the data and cleaning procedures are provided in Appendices B and D.

and carriers, and is calibrated targeting the aggregate share of domestic inputs in the economy. $\mu_\zeta$ is normalized to one so that $\mu_\varphi$ is interpretable as the relative mean productivity. The variance $\sigma^2_\alpha$ is calibrated targeting the average dispersion and average maximum of $s_{ijmt}$ across importers, the average dispersion and average maximum of $x_{ijmt}$ across carriers, and the correlation between the two.

**Home bias and price of imports.** We normalize $\eta = 0.5$ without loss of generality. We calibrate the factory-gate price of imports $\overline{p}_{iF}$, assumed common across importers, by targeting two moments jointly: the average share of transportation costs in the price of imported goods, $s_{i\tau}$, and the aggregate share of domestic inputs in the economy.

**Algorithm for Estimating the Model** We estimate the model parameters using simulated method of moments (SMM). For a given parameter vector, we simulate 50 economies, solve for the equilibrium of each, compute model moments for each simulated economy, and compare the average simulated moments to their empirical counterparts. Each simulated economy is solved numerically using a nested fixed-point (NFXP) algorithm. Given initial values of $P$ and $f$, we solve an inner fixed-point problem to recover the bilateral freight prices $\tau_{ij}$ satisfying the Nash-in-Nash bargaining conditions in Equation (5). Given the resulting $\tau_{ij}$, we update the outer fixed point over $P$ and $f$ until both inner and outer fixed-point conditions are jointly satisfied.[23]

[23]The fixed cost of importing $f$ is determined separately for each simulated economy by finding the value that makes the marginal non-importing firm indifferent between domestic sourcing and importing. Specifically, we identify the most productive non-importing firm in equilibrium, indexed by $i^\star$, compute its counterfactual profit $\pi^{\mathrm{CF}}_{i^\star}$ under importing, and recover $f$ from: $f = \left(\pi^{\mathrm{CF}}_{i^\star} - \pi^{\mathrm{B}}_{i^\star}\right)/w$, where $\pi^{\mathrm{B}}_{i^\star}$ is the baseline profit from not importing.

The parameter vector $\Theta = \{\overline{p}_{iF},\, \sigma^2_\varphi,\, \mu_\varphi,\, \sigma^2_\zeta,\, \sigma^2_\alpha\}$ is estimated via two-step SMM:

$$\widehat{\Theta}_{\text{SMM}} = \arg\min_{\Theta} \left(\frac{\hat{m}(\tilde{x} \mid \Theta) - m(x)}{m(x)}\right)' W \left(\frac{\hat{m}(\tilde{x} \mid \Theta) - m(x)}{m(x)}\right), \tag{16}$$

where $m(x)$ is the vector of empirical moments, $\hat{m}(\tilde{x} \mid \Theta)$ is the vector of average simulated moments, and $W$ is the optimal weighting matrix estimated in the first step. Given the nonlinearity of the model, we use simulated annealing as the optimization routine in both steps, declaring convergence when the objective function shows no improvement exceeding a minimum threshold for more than 600 consecutive evaluations.

**Estimated Parameters and Model Fit** Panels A and B of Table 8 report the estimated parameters with standard errors and the data and simulated moments, respectively. The model fits the targeted moments well and the parameters are precisely estimated. The model reproduces the aggregate domestic input share and the average share of transportation costs in the price of imports, $s_{i\tau}$, with an estimated factory-gate import price $\overline{p}_{iF} = 1.53$. The estimated productivity distributions imply that carriers are on average more efficient than domestic firms, which lowers transportation costs and drives the relative price of domestic and imported inputs, and through it the aggregate domestic share and $s_{i\tau}$. The model also reproduces the dispersion in domestic sales and in carriers' aggregate market shares, which identify the variances $\sigma^2_\varphi$ and $\sigma^2_\zeta$, respectively. The variance of idiosyncratic importer-carrier taste shocks $\sigma^2_\alpha$, jointly with the productivity variances, targets the distribution of bilateral shares: the model captures the average maximum of $s_{ijmt}$, its dispersion, and the correlation between $s_{ijmt}$ and $x_{ijmt}$ well, but understates the average maximum of $x_{ijmt}$ across carriers and its dispersion.

**On Identification** Structural estimation via SMM inevitably raises questions of identification, particularly in models where analytical identification arguments are unavailable. We address this through two approaches standard in the literature. First, we follow Andrews et al. (2017) and report the sensitivity matrix $\Lambda = -(G'WG)^{-1}G'W$ in Table H.1 in Appendix H, where $W$ is the probability limit of $\widehat{W}$ and $G$ is the Jacobian of the probability limit of $\hat{m}(\tilde{x} \mid \Theta)$ evaluated at $\Theta_0$. This matrix provides a local linear approximation of the mapping from moments to parameters and quantifies how sensitively each parameter responds to each moment. Second, we examine how variation in individual parameters affects simulated moments, holding all other parameters fixed (Kaplan, 2012; Berger and Vavra, 2015; Morten, 2019). Identification requires that each parameter predominantly influences a distinct subset of moments. Parameters that move different moments in different directions are well-identified even if no single moment is exclusively informative about one parameter. Figure H.1 in Appendix H plots the percentage change in each simulated moment in response to variation in each estimated parameter. The patterns confirm that the moments respond differentially across parameters, supporting the identification of the full parameter vector.

Table 8: Parametrization

| **A. Estimated parameters** | **Estimate** | **(Std. Error)** |
|---|---|---|
| Std of domestic firm productivity, $\sigma_{z_i}$ | 0.23 | (0.01) |
| Mean of domestic firm productivity, $\mu_{z_i}$ | -0.08 | (0.03) |
| Std of carrier productivity, $\sigma_\zeta$ | 1.03 | (0.08) |
| Std idiosyncratic importer-carrier match shock, $\sigma_{\alpha_{ij}}$ | 0.23 | (0.01) |
| Factory-gate price of imports, $\overline{p_{iF}}$ | 1.53 | (0.07) |
| Fixed cost of importing, $f$ | 0.07 | (0.00) |
| **B. Targeted moments** | **Data** | **Model** |
| Std sales share in domestic economy | 0.03 | 0.04 |
| Aggregate domestic share | 0.89 | 0.96 |
| Correlation$(s_{ij}, x_{ij})$ | 0.14 | 0.13 |
| Average Max $x_{ij}$ | 0.49 | 0.32 |
| Median Std in $x_{ij}$ across carriers | 0.19 | 0.09 |
| Average Max $s_{ij}$ | 0.43 | 0.49 |
| Median Std in $s_{ij}$ across importers | 0.17 | 0.16 |
| Average $s_{i\tau}$ | 0.10 | 0.09 |
| Std aggregate shares across carriers | 0.22 | 0.19 |

**Notes**: Panel A reports parameters estimated via two-step SMM with corresponding standard errors. Panel B reports the targeted moments in the data and in the simulated model. Simulated moments are averaged across 50 economies. Data moments are computed using Chilean customs transaction data from 2007 to 2022 and the Survey of Manufacturing Industries (ENIA) from 1995 to 2018. Additional details on the data and cleaning procedures are provided in Appendices B and D.

## 4.3 The Aggregate Impact of Endogenous Trade Costs

We examine how the endogeneity of transportation prices, arising from bargaining frictions and decreasing returns to scale in the transportation sector, shapes equilibrium prices and the transmission of tariff and cost shocks.

### 4.3.1 Bargaining and Equilibrium Transportation Prices

In this set of results, we compare the price level, and allocations of shares, in our baseline specification with alternative scenarios in which remove some features of our model. Table 9 shows that both bargaining power and decreasing returns to scale are crucial for explaining price levels and allocations in the transportation sector. Holding the returns to scale parameter fixed, an economy that abstracts from importer bargaining power ($\phi \to 0$) exhibits transportation prices that are on average 46 percent higher than in the baseline (0.35 vs. 0.24), as carriers can extract larger markups unconstrained by buyer bargaining power. Higher transportation costs raise import prices, which in turn increase the marginal costs of domestic producers, resulting in consumer prices that are 0.13 percentage points above the baseline.[24]

Imposing constant returns to scale ($\theta \to 1$) while holding bargaining power fixed generates qualitatively similar effects on prices, but through a fundamentally different mechanism

[24]The same chain of transmission operates in reverse when bargaining power is fully assigned to importers ($\phi \to 1$), leading to lower transportation prices, lower import prices, and lower consumer prices.

Table 9: Comparative Statics

| | $\tau_{ij}$ | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Weighted Mean | Mean | Median | $\max s_{ij}$ | $\max x_{ij}$ | $s_{i\tau}$ | $s_{iF}$ | % Change in CPI level |
| Baseline | 0.24 | 0.27 | 0.24 | 0.49 | 0.32 | 0.09 | 0.04 | - |
| $\phi \to 0$ | 0.35 | 0.38 | 0.34 | 0.46 | 0.31 | 0.13 | 0.04 | 0.13 |
| $\phi \to 1$ | 0.21 | 0.24 | 0.21 | 0.51 | 0.34 | 0.09 | 0.04 | -0.04 |
| $\theta \to 1$ | 0.68 | 1.56 | 0.97 | 0.73 | 0.32 | 0.24 | 0.03 | 0.40 |
| $\theta \to 1 \& \phi \to 0$ | 1.32 | 2.20 | 1.49 | 0.60 | 0.31 | 0.35 | 0.02 | 0.68 |

**Notes**: The table reports equilibrium objects for alternative values of importer bargaining power $\phi$ and carrier returns to scale $\theta$. Reported objects include the weighted mean, mean, and median of bilateral transportation prices, the maximum bilateral shares $s_{ij}$ and $x_{ij}$, the average share of transportation costs in the price of imported goods, the share of foreign intermediates across importers; and the percent difference in the domestic price index relative to the baseline. Weights are bilateral transportation quantities within each market-time cell.

with empirically implausible implications for market allocations. Under constant returns, each carrier's marginal cost is flat rather than increasing in quantity supplied. This eliminates the mechanism by which importers exert buyer power: since carriers no longer face rising costs at the margin, importers cannot extract lower prices and transportation prices therefore rise. At the same time, more productive carriers maintain a permanent cost advantage that is never eroded by decreasing returns, allowing them to capture disproportionately large market shares. The resulting allocation is highly concentrated and at odds with the data: the largest carriers reach bilateral shares $s_{ij}$ of approximately 73 percent, substantially above both the data and the baseline model.

#### 4.3.2 Tariff shocks and Gains from Trade

We assess whether endogenous transportation costs affect the welfare effects of tariff shocks. We compare our results against a standard iceberg trade cost model and derive a first-order approximation of the welfare effects of tariff shocks. We show that the pass-through of tariffs to consumer prices is approximately 50 percent higher under iceberg trade frictions than in our model, driven primarily by the presence of decreasing returns to scale in the transportation sector.

**Approximation for the gains from trade** We model a rise in ad valorem tariffs $\bar{t}$ as an increase in the factory-gate price of imported goods from $\overline{p}_F$ to $\overline{p}_F(1+\bar{t})$. We measure aggregate welfare as changes in the aggregate consumer price index $P$. We can approximate the change in the aggregate consumers' price index following a change in ad valorem tariffs as follows:

$$\frac{\partial \log P}{\partial \log \bar{t}} \approx \frac{\beta \sum_i s_i s_{iF} \left(1 - s_{i\tau} + s_{i\tau} \cdot \frac{\partial \log \tau_i}{\partial \log \bar{t}}\right)}{1 - \beta \sum_i s_i (1 - s_{iF})} \equiv \frac{\beta \cdot (\mathbf{s} \circ \mathbf{s}_F)^\top (\mathbf{1_I} - \mathbf{s}_\tau + \mathbf{s}_\tau \circ \mathbf{g})}{1 - \beta \cdot \mathbf{s}^\top (\mathbf{1_I} - \mathbf{s}_F)} \tag{17}$$

where $s_i$ is firm $i$'s share of final consumption, $s_{iF}$ is its share of foreign intermediates, $s_{i\tau}$ is the share of transportation costs in its import price, and $\frac{\partial \log \tau_i}{\partial \log \bar{t}}$ is the endogenous response of transportation costs to the tariff shock. The right-hand side rewrites the expression in matrix form, with $\circ$ denoting the Hadamard product and $\mathbf{1_I}$ a vector of ones. The key object is the vector $\mathbf{g} = \left(\frac{\partial \log \tau_1}{\partial \log \bar{t}}, \dots, \frac{\partial \log \tau_I}{\partial \log \bar{t}}\right)^\top$, which can be approximated as:

$$\mathbf{g} = -\left[\mathbf{I} + \xi \frac{1-\theta}{\theta} \mathbf{B} \ \mathrm{diag}(\mathbf{s}_\tau)\right]^{-1} \left(\xi \frac{1-\theta}{\theta} \mathbf{B}(\mathbf{1} - \mathbf{s}_\tau)\right), \tag{18}$$

where $\mathbf{s}_\tau = (s_{1\tau}, \dots, s_{I\tau})^\top$, and $B_{ii'} = \sum_j s_{ij} \, \Phi_{ij} \, x_{i'j}$. The pass-through factor $\Phi_{ij} \equiv \frac{1}{1+\Gamma_{ij}+\Lambda_{ij}}$ summarizes the attenuation of the tariff shock due to bilateral markup adjustments $\Gamma_{ij}$ and the cost elasticity $\Lambda_{ij}$ (formal expressions are provided in Appendix A.4. Appendix A.4 provides the full derivation of Equations (17) and (18). Figure H.4 in Appendix A.4 confirms that the approximation tracks the true welfare change closely across the range of tariff shocks considered.

**Baseline vs. Iceberg** Table 10 shows that endogenous transportation costs reduce the welfare costs of tariff shocks relative to the iceberg case. A 10% ad valorem tariff increase raises the aggregate price index by 0.21% in our baseline model, abstracting from firms' entry and exit. Under standard iceberg trade costs, the same tariff raises the price index by 0.32%, approximately 52% more than in the baseline.

The lower welfare cost in the baseline is driven by a decline in transportation costs that partially offsets the rise in the factory-gate price of imports. Formally, the total cost of imported goods for importer $i$ changes as $\overline{p}_{iF}(1+\bar{t})$ rises while $\tau_i$ falls, so the net increase in import costs is attenuated. This attenuation is captured by the term in parentheses in the numerator of Equation (17), which equals one under iceberg costs and is strictly less than one in the baseline since $\frac{\partial \log \tau_i}{\partial \log \bar{t}} < 0$.[25]

The decline in transportation costs is primarily driven by carriers' cost-side adjustments. Higher import prices reduce demand for imported goods and, given the Leontief structure of production, lower demand for transportation services. Under decreasing returns to scale, this contraction in demand reduces transportation prices. The term $\xi \frac{1-\theta}{\theta} \mathbf{B}(\mathbf{1} - \mathbf{s}_\tau)$ in Equation (18) captures this channel under the assumption of fixed markups and cost elasticity ($\Phi_{ij} = 1$). As shown in the first bar of Figure 3, this returns-to-scale channel is the dominant driver of overall pass-through attenuation.

Markup adjustments are comparatively small and primarily reflect strategic complementarities. Variable markups and non-zero cost elasticities both generate incomplete pass-through ($\Phi_{ij} \leq 1$). Carriers' oligopolistic markups dampen pass-through through strategic complementarities, while importers' markdown adjustments amplify it through strategic substitutability (Amiti et al., 2019a; Alviarez et al., 2026). The second and third bar of Figure 3 show the respective contributions of markup and markdown adjustments; the former dominates, resulting in a more muted decline in transportation costs.[26]

[25] A small part of the difference between the two welfare effects reflects the additive structure of transportation costs. Additivity implies that the pass-through of the tariff to import prices $p_{iF}$ equals $(1 - s_{i\tau})$, whereas it equals one in the multiplicative iceberg case.

[26] The fourth bar reports the contribution of cost elasticity, capturing additional incomplete pass-through from endogenous cost adjustments beyond the initial transportation price shock.

Table 10: Pass-through of Tariffs

| Model | Description | Bargaining | RTS | Aggregate PT |
|---|---|---|---|---|
| (1) | Baseline | Yes | Yes | 0.21 |
| (2) | Iceberg | No | No | 0.32 |
| (3) | Additive | No | No | 0.29 |
| (4) | $\phi = 0$ | No | Yes | 0.17 |
| (5) | $\phi = 1$ | Yes | Yes | 0.23 |

**Notes**: The table displays the percentage change in welfare due to a 10 percent tariff increase for different values of importer bargaining power, $\phi$, and carrier's return to scale, $\theta$. Welfare is measured as the consumer aggregate price index. Numbers are averaged across simulated economies.

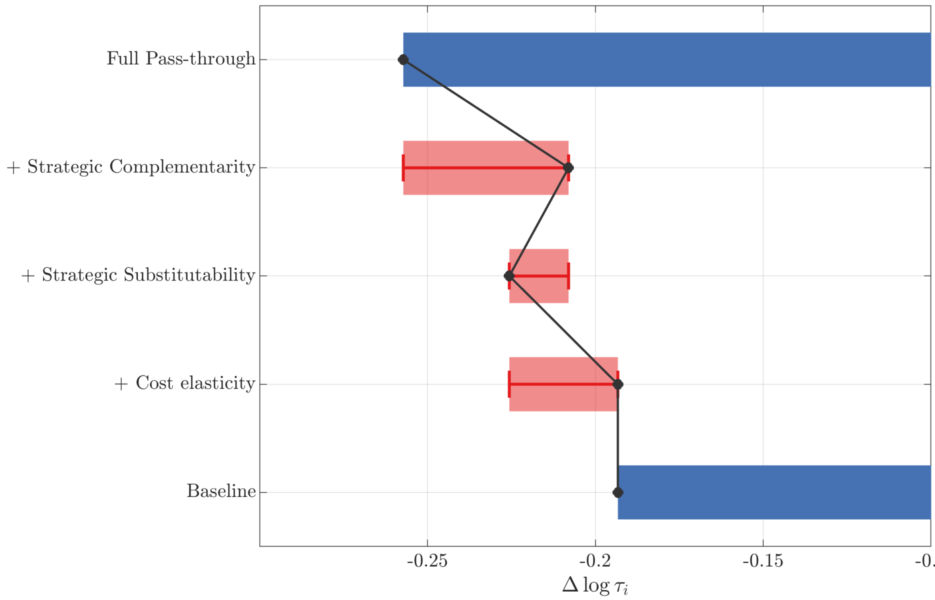
The welfare effects of tariffs are heterogeneous across importers due to differences in bargaining positions. Figure H.2 in Appendix F shows that the increase in bilateral markups is more pronounced for importer-carrier pairs with higher bilateral shares $s_{ij}$, reflecting greater exposure to carrier market power. Importers with larger buyer shares experience smaller markup increases, reflecting their stronger bargaining position. Although the tariff is uniform, its impact on transportation costs and total import costs is therefore highly heterogeneous across importers.

**Entry and exit** Figure H.3 in Appendix F shows that accounting for the entry and exit of importers magnifies the welfare losses from tariffs in a non-linear manner, with the amplification increasing in the size of the tariff shock. Tariffs raise the price of imported inputs, reducing the incentive for domestic firms to import. When a firm exits the importing market, its marginal cost increases, raising its output price and contributing to a higher aggregate price index. The extensive margin therefore amplifies the negative welfare effects of tariffs.

At the same time, the exit of importers reshapes competition in the transportation market by reallocating market shares toward surviving firms. As surviving importers account for a larger share of each carrier's business on average, that is, $x_{ij}$ rises, they exercise stronger buyer market power and negotiate lower transportation prices. This reflects a form of strategic substitutability across importers: as some firms exit, carriers' outside options deteriorate, raising the marginal value of the remaining importers and strengthening their bargaining position. Lower transportation costs for surviving firms partially offset the increase in input prices caused by the tariff.

**Effect of bargaining and returns to scale** Tariff pass-through is primarily governed by returns to scale, whereas bargaining power matters indirectly through its effect on equilibrium allocations. Table 10 reports the pass-through of tariffs to the consumer price index under alternative model specifications for bargaining power and returns to scale. Shifting bargaining power to one side of the market alters both the nature of strategic interactions, favoring either complementarities or substitutability, and the equilibrium expenditure shares (see Table 9). For example, when all bargaining power is allocated to carriers ($\phi \to 0$), bilateral markup adjustments reflect only strategic complementarities. In the absence of offsetting substitution

Figure 3: Decomposition




**Notes**: The figure displays the decomposition of the numerator term in Equation (18) for a 10% ad valorem tariff, averaged across simulated economies. The first bar reports the baseline returns-to-scale channel, setting $\Phi_{ij} \equiv 1$, which isolates the term $\xi \frac{1-\theta}{\theta} \mathbf{B}(\mathbf{1} - \mathbf{s}_\tau)$. The second bar adds the contribution of strategic complementarities by setting $\Phi_{ij} = 1/(1 + \frac{\partial \log \overline{\overline{\mu}}_{ij}}{\partial \log \tau_{ij}})$. The third bar further adds the contribution of strategic substitutability by setting $\Phi_{ij} = \frac{1}{1+\Gamma_{ij}}$. The fourth bar adds the contribution of cost elasticity by setting $\Phi_{ij} = \frac{1}{1+\Gamma_{ij}+\Lambda_{ij}}$. The last bar reports the full effect combining all channels.

effects, transportation prices respond less to tariff shocks, resulting in higher pass-through to consumer prices. However, under this specification, the share of transport costs ($s_{i\tau}$) is higher, while the share of imported inputs ($s_{iF}$) is lower than in the baseline model, both of which dampen the overall pass-through to consumer prices. As Table 10 shows, this second effect dominates, leading to a lower welfare cost of tariffs relative to the baseline. The opposite holds when bargaining power is entirely shifted to importers ($\phi \to 1$).[27]

**Importer-exporter bargaining and PT into factory gate prices** An important consideration in interpreting our results concerns the scope of the tariff transmission mechanism we analyze. In our framework, tariffs apply to the exogenously given factory-gate price, $\overline{p_{iF}}$, namely the price paid by the importer to the exporter for the imported good. Our analysis isolates the effects of tariffs operating through equilibrium adjustments in the transportation market, conditional on a given change in factory-gate prices.[28] The importer-carrier bargaining margin studied in this paper is therefore complementary to, rather than a substitute for,

[27]Assuming constant returns to scale in the transportation sector lowers the welfare cost of tariffs by 20 percent (0.17 vs 0.21) relative to the baseline. In this scenario, transportation prices are unresponsive to tariff shocks because carriers face constant marginal costs, implying that the term $\frac{\partial \log \tau_i}{\partial \log t}$ in Equation (17) is zero. Consequently, the deviation from the iceberg benchmark is not driven by the endogeneity of transportation costs, but rather by equilibrium differences in expenditure shares: specifically, a higher share of transport costs ($s_{i\tau}$) and a lower share of imported inputs in production ($s_{iF}$). See Table H.2 in Appendix H.

[28]An interesting avenue for future research would be to endogenize the joint determination of export and transportation prices.

the exporter-importer bargaining margin in Alviarez et al. (2026). The two relationships operate at different stages of the supply chain, over the factory-gate price and the freight rate, respectively, and jointly determine the total landed cost faced by domestic producers (Equation 15). As a result, the mechanism we study should be interpreted as an additional margin through which trade policy affects import costs and allocation outcomes. More broadly, we remain agnostic about the determinants of tariff pass-through into factory-gate import prices, which are the subject of a large and growing literature (Amiti et al., 2019b; Fajgelbaum et al., 2020; Ganapati and Hottman, 2026). Rather than taking a stand on the extent to which foreign exporters absorb tariffs through lower producer prices, our analysis conditions on observed changes in factory-gate prices and studies how these changes propagate through the transportation sector. In this sense, our mechanism is orthogonal to standard pass-through channels operating at the exporter-importer margin and can be combined with alternative models of pricing-to-market, imperfect competition, or exporter bargaining.

#### 4.3.3 Cost Shock to the Transportation Sector

We show how symmetric cost shocks, such as oil price increases and carbon taxes, and asymmetric shocks, such as the United States Trade Representative (USTR) remedies, propagate through transportation prices to aggregate welfare.

**Symmetric cost shocks: oil shock and carbon tax** We assess the aggregate welfare effects of cost shocks affecting all carriers simultaneously, using an oil price shock and a global carbon tax as leading examples.

Fuel costs represent the single largest component of shipping costs, accounting for approximately 47% of the total (Stopford, 2008). Large fluctuations in oil prices, such as the spike in 2022, can therefore significantly affect carriers' costs, transportation prices, and ultimately consumer prices. We model a severe oil price shock as a 100% increase in the price of oil, which translates into a 47% increase in carriers' marginal costs.[29]

Figure 4 shows that this cost shock raises transportation prices by approximately 30%, implying a pass-through rate of carrier marginal costs to transportation prices of around 0.3, consistent with estimates in Hummels (2007) and slightly above Brancaccio et al. (2023). Despite the large cost increase, the aggregate consumer price index rises by less 1%, reflecting the combination of incomplete pass-through and the small share of transportation costs in the consumer price index.

Turning to the carbon tax, the international shipping sector emits approximately 800 million tonnes of $CO_2$ annually, representing around 3% of global greenhouse gas emissions (Lugovskyy et al., 2023). We model a carbon tax of €50 per tonne of $CO_2$, consistent with EU Commission projections for the 2030s, which translates into an approximately 5% increase in

[29]The price of crude oil (Brent) peaked at approximately $120 per barrel between March and June 2022, up from a pre-pandemic average of around $60. Figure H.5 in Appendix F documents this episode.

carriers' marginal costs.[30] Formally, the carbon tax enters as a proportional increase in carrier marginal costs: $\tau_{ij} = \mu_{ij} c_j (1 + t^{green})$, where $t^{green} \approx 0.05$.

Figure 4 shows that the carbon tax is passed through incompletely to transportation prices, which rise by approximately 4%, and that the resulting increase in the aggregate consumer price index is negligible (Coster et al., 2024). The incomplete pass-through reflects two mechanisms: as the share of transportation costs in import prices $s_{i\tau}$ rises, carriers face a higher perceived demand elasticity, which compresses their markups and limits their ability to pass the tax forward, a secondary role is played by decreasing returns to scale, which dampens the cost response. The small consumer welfare cost should be interpreted as an upper bound, since we abstract from any benefits to consumers from reduced carbon emissions.

**Asymmetric cost shocks: USTR remedies** We use our model to study the potential impact of the asymmetric policies proposed under the USTR's Section 301 investigation into Chinese dominance in the maritime sector. The USTR report documents that China aggressively targeted the maritime sector and proposes several remedies, including a service fee of \$1 million imposed on Chinese vessel operators entering a U.S. port.[31] The World Shipping Council (WSC) testified that such remedies could increase costs per container by \$750, representing an increase of between 30% and 100% relative to current spot prices.

We model this scenario as a cost increase applied to a subset of carriers, following Chen (2024c).[32] We focus on a 30% cost increase, the lower bound of the WSC's estimates.

Directly affected carriers raise their transportation prices by approximately 10%, passing through the cost increase incompletely. The cost shock erodes their competitiveness, causing them to lose market shares and, with it, market power, which further compresses their markups and profits. Unaffected carriers benefit from the resulting competitive advantage: their seller shares $s_{ij}$ rise, strengthening their market power over importers and allowing them to raise markups and prices by approximately 5%, generating additional profits. As in the previous cases, the aggregate consumer price index rises only marginally, reflecting the small share of transportation costs in final goods prices.

# 5 Conclusion

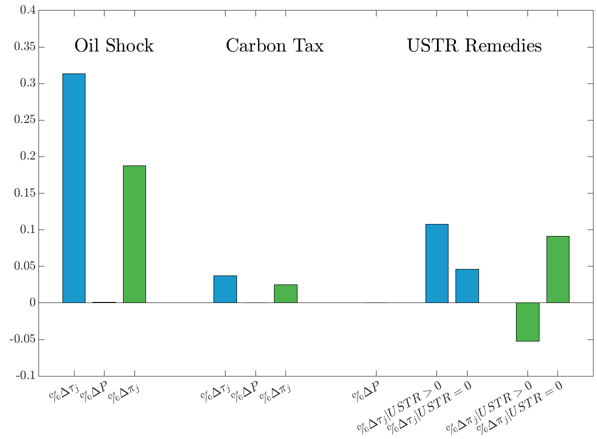
This paper examines the role of imperfect competition and bilateral negotiations in the transportation sector and their impacts on international trade. Our analysis provides several key

[30] The €50 benchmark lies between the €100 per tonne used in Coster et al. (2024) and the €29 per tonne in Shapiro (2016). We compute the carbon cost share of Chilean imports by: (1) computing total tonne-km for each transport mode using customs and BACI/CEPII data; (2) applying mode-specific $CO_2$ emission coefficients from ECTA/CEFIC (7 $gCO_2$/tonne-km for sea, 602 $gCO_2$/tonne-km for air, and 62 $gCO_2$/tonne-km for road); and (3) computing the implied carbon tax cost relative to total freight costs.

[31] The report and testimony are available at: https://ustr.gov/sites/default/files/enforcement/301Investigations/FRNActionabilityChinaTargetingMaritime.pdf and https://www.worldshipping.org/s/Hearing-Testimony_World-Shipping-Council_Joe-Kramek-USTR-2025-0003-filed-20-March.pdf.

[32] Specifically, we apply the additional cost to one carrier at a time and report average effects across iterations.

Figure 4: Oil Shock, Carbon Tax, and USTR Remedies




**Notes**: The figure reports changes in transportation costs (blue), the domestic price index (red), and carriers' profits (green) across three counterfactual scenarios: a symmetric increase in fuel costs from an oil shock (left), a carbon tax (center), and an asymmetric increase in costs from USTR port fees (right). Entry and exit of importers is held fixed in all three exercises.

contributions to the literature on international trade and industrial organization.

Using detailed Chilean customs data, we document empirical evidence of high concentration in the transportation sector. We also provide evidence of bilateral negotiations and dual market power between carriers and importers in determining transportation prices. These findings challenge the common assumption in trade literature of perfectly competitive transportation markets and reject the ”iceberg” cost assumption used in many trade models.

# A Derivations and Proofs

## A.1 Derivations of Bilateral Prices

This appendix derives the solution for bilateral transportation prices, $\tau_{ij}$, within the framework described in Section 3, following previous work from Alviarez et al. (2026).

**Importer** Given the assumptions in Section 3, importer $i$'s profits in case of successful negotiations can be written as:

$$\pi_i = (\mu_i - 1)\mu_i^{-\sigma} u_i^{1-\sigma}, \tag{19}$$

where $\mu_i$ is constant and $u_i$ is the marginal cost of importer $i$. It follows that the derivative of $i$'s profits with respect to the bilateral transportation price $\tau_{ij}$ is:

$$\begin{aligned}
\frac{\partial \pi_i}{\partial \tau_{ij}} &= (\mu_i - 1)\mu_i^{-\sigma}(1-\sigma)u_i^{-\sigma}\frac{\partial u_i}{\partial \tau_{ij}} \\
&= (\mu_i - 1)\mu_i^{-\sigma}(1-\sigma)u_i^{-\sigma}\frac{\partial u_i}{\partial p_{iF}}\frac{\partial p_{iF}}{\partial \tau_i}\frac{\partial \tau_i}{\partial \tau_{ij}} \\
&= (\mu_i - 1)\mu_i^{-\sigma}(1-\sigma)u_i^{-\sigma}\gamma\frac{u_i}{p_{iF}}\frac{\tau_{ij}^{-\rho}}{\tau_i^{-\rho}} \\
&= (\mu_i - 1)p_i^{-\sigma}(1-\sigma)\frac{q_{iF}}{q_i}\frac{t_{ij}}{t_i} \\
&= (\mu_i - 1)(1-\sigma)t_{ij}
\end{aligned}$$

where the last equation is obtained noticing that $t_i = q_{iF}$ given the Leontief production function.

We can derive importer $i$'s profits in case of failed negotiation, $\pi_{i(-j)}$, provided that the cost of a unit of transportation bundle without $j$ is now:

$$\widetilde{\tau}_i = \tau_i(1 - s_{ij})^{\frac{1}{1-\rho}} = \tau_i(1 + \Delta\tau), \tag{20}$$

where $s_{ij} = \frac{\tau_{ij} t_{ij}}{\sum_{z \in J_i} \tau_{iz} t_{iz}}$ is the share of carrier $j$ in total transportation costs of importer $i$. Thus, we can write:

$$\pi_{i(-j)} = (\mu_i - 1)\mu_i^{-\sigma}\widetilde{u}_i^{1-\sigma} = (\mu_i - 1)q_i u_i \left(1 + s_{i\tau}\Delta\tau\right)^{\gamma(1-\sigma)},$$

where $s_{i\tau} = \frac{\tau_i}{p_{iF}} = \frac{\tau_i}{\overline{p_{iF}} + \tau_i}$ is the share of transportation costs in the price of imported goods. It follows immediately that the gains from trade for importer $i$ are:

$$\Omega_{ij} \equiv \pi_i(\tau_{ij}) - \pi_{i(-j)} = (\mu_i - 1)q_i u_i \left(1 - (1 + s_{i\tau}\Delta\tau)^{\gamma(1-\sigma)}\right). \tag{21}$$

**Carrier** By the same token, we derive the gains from trade for carrier $j$. The profits of carrier $j$ in case of successful negotiation are:

$$\pi_j(\tau_{ij}) = \tau_{ij}t_{ij} + \sum_{z \neq i \in Z_j} \tau_{zj}t_{zj} - \theta c_j t_j, \tag{22}$$

where $c_j$ is the marginal cost of production given the upward-slope supply curve, and $t_j = \sum_{i \in Z_j} t_{ij}$. It is immediate to show that:

$$\begin{aligned}
\frac{\partial \pi_j(\tau_{ij})}{\partial \tau_{ij}} &= t_{ij} + \tau_{ij}\frac{\partial t_{ij}}{\partial \tau_{ij}} - \theta t_j \frac{\partial c_j}{\partial \tau_{ij}} - \theta c_j \frac{\partial t_j}{\partial \tau_{ij}} \\
&= t_{ij} + \tau_{ij}\frac{\partial t_{ij}}{\partial \tau_{ij}} - c_j \frac{\partial t_{ij}}{\partial \tau_{ij}} \\
&= t_{ij}\left(1 - \epsilon_{ij} + \epsilon_{ij}\frac{c_j}{\tau_{ij}}\right),
\end{aligned}$$

where $\epsilon_{ij} = -\frac{\partial t_{ij}}{\partial \tau_{ij}}\frac{\tau_{ij}}{t_{ij}}$ is the perceived demand elasticity of the carrier. Specifically, given the structure on the importer side,

$$\epsilon_{ij} = (1 - s_{ij}) \cdot \rho + s_{ij} \cdot (s_{i\tau} \cdot (1 - \gamma + \sigma \cdot \gamma)) \tag{23}$$

Moreover, in case of failed negotiations, the profits of carrier $j$ become:

$$\pi_{j(-i)} = \sum_{z \neq i \in Z_j} \tau_{zj}t_{zj} - \theta \widetilde{c_j} \sum_{z \neq i \in Z_j} t_{zj} = \sum_{z \neq i \in Z_j} \tau_{zj}t_{zj} - \theta \widetilde{c_j} t_j (1 - x_{ij}), \tag{24}$$

with $\widetilde{c_j} = c_j\,(1 - x_{ij})^{\frac{1-\theta}{\theta}}$, where $x_{ij} = \frac{t_{ij}}{t_j}$ is the share of total sales of $j$ purchased by importer $i$.

Combining the equations above, we can write the gains from trade for carrier $j$ as:

$$\Psi_{ij} \equiv \pi_j(\tau_{ij}) - \pi_{j(-i)} = \tau_{ij}t_{ij} - \theta c_j t_j \left[1 - (1 - x_{ij})^{\frac{1}{\theta}}\right] = t_{ij}(\tau_{ij} - c_j \widehat{\mu_{ij}}), \tag{25}$$

where $\widehat{\mu_{ij}} = \theta \frac{1 - (1 - x_{ij})^{\frac{1}{\theta}}}{x_{ij}}$ is the markup in the oligopsony case.

**Bilateral prices** Given the expressions for the gains from trade above, the FOC for the problem in Equation (5) is:

$$0 = \frac{\partial \pi_j}{\partial \tau_{ij}} + \overline{\phi}\frac{\pi_j - \pi_{j(-i)}}{\pi_i - \pi_{i(-j)}}\frac{\partial \pi_i}{\partial \tau_{ij}},$$

where $\overline{\phi} = \frac{\phi}{1-\phi}$. Substituting the relevant expressions from above, we get:

$$\begin{aligned}
0 &= (1 - \epsilon_{ij} + \epsilon_{ij}\frac{c_j}{\tau_{ij}}) + \overline{\phi}\frac{\tau_{ij} - c_j\widehat{\mu_{ij}}}{q_i u_i\,(1 - (1 + s_{i\tau}\Delta\tau)^{\gamma(1-\sigma)})}(1 - \sigma)t_{ij} \\
&= -1 + \frac{\epsilon_{ij}}{\epsilon_{ij} - 1}\frac{c_j}{\tau_{ij}} - \overline{\phi}\frac{c_j}{\tau_{ij}}\widehat{\mu_{ij}}\frac{1 - \sigma}{\epsilon_{ij} - 1}\frac{\tau_{ij}t_{ij}(\mu_i - 1)}{\Omega_{ij}} + \overline{\phi}\frac{1 - \sigma}{\epsilon_{ij} - 1}\frac{\tau_{ij}t_{ij}(\mu_i - 1)}{\Omega_{ij}}
\end{aligned}$$

$$= -1 + \overline{\overline{\mu_{ij}}}\frac{c_j}{\tau_{ij}} + \bar{\phi}\lambda_{ij}\widehat{\mu_{ij}}\frac{c_j}{\tau_{ij}} - \bar{\phi}\lambda_{ij}$$
$$\tau_{ij} = c_j\left(\omega_{ij}\widehat{\mu_{ij}} + (1-\omega_{ij})\overline{\overline{\mu_{ij}}}\right).$$

which is Equation (6) in the main text, where $\omega_{ij} = \frac{\bar{\phi}\lambda_{ij}}{1+\bar{\phi}\lambda_{ij}}$, $\lambda_{ij} = \frac{\sigma-1}{\epsilon_{ij}-1}\frac{\tau_{ij}t_{ij}(\mu_i-1)}{\Omega_{ij}}$, and $\overline{\overline{\mu_{ij}}} = \frac{\epsilon_{ij}}{\epsilon_{ij}-1}$ is the standard markup in case of oligopoly.

## A.2 Derivations of Bilateral Prices in Quantitative Model

Solution for bilateral transportation prices.

Given $t_i = \left(\sum_{j\in J_i}\alpha_{ij}^{\frac{1}{\rho}}t_{ij}^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}}$ and $\tau_i = \left(\sum_{j\in J_i}\alpha_{ij}\tau_{ij}^{1-\rho}\right)^{\frac{1}{1-\rho}}$:[33]

1. Define the failed negotiation bit for the importer in Nash-bargaining:

$$\pi_{i(-j)} = (\mu_i - 1)\mu_i^{-\sigma}\widetilde{u_i}^{1-\sigma}P^\sigma Y$$
$$\widetilde{u_i} = w^{1-\beta}p_x^\beta = w^{1-\beta}\left(\eta^\xi p_D^{1-\xi} + (1-\eta)^\xi(\alpha_{iq}\bar{p_F} + \alpha_{it}\widetilde{\tau_i})^{1-\xi}\right)^{\frac{\beta}{1-\xi}}$$

where $\widetilde{\tau_i} = \tau_i(1-s_{ij})^{\frac{1}{1-\rho}} = \tau_i(1+\Delta\tau)$, with $s_{ij} = \alpha_{ij}\left(\frac{\tau_{ij}}{\tau_i}\right)^{1-\rho}$. We can therefore rewrite $\widetilde{u_i}$ as follows:

$$\begin{aligned}\widetilde{u_i} &= w^{1-\beta}p_x^\beta = w^{1-\beta}\left(\eta^\xi p_D^{1-\xi} + (1-\eta)^\xi(\alpha_{iq}\bar{p_F} + \alpha_{it}\widetilde{\tau_i})^{1-\xi}\right)^{\frac{\beta}{1-\xi}}\\ &= w^{1-\beta}\left(\eta^\xi p_D^{1-\xi} + (1-\eta)^\xi[p_{iF}(1+s_{i\tau}\Delta\tau)]^{1-\xi}\right)^{\frac{\beta}{1-\xi}}\\ &= w^{1-\beta}p_x^\beta\left(1+s_i^F[(1+s_{i\tau}\Delta\tau)^{1-\xi}-1]\right)^{\frac{\beta}{1-\xi}}\\ &= u_i\left(1+s_i^F[(1+s_{i\tau}\Delta\tau)^{1-\xi}-1]\right)^{\frac{\beta}{1-\xi}}\end{aligned}$$

where $s_i^\tau = \frac{\alpha_{it}\tau_i}{\alpha_{iq}\bar{p_F}+\alpha_{it}\tau_i}$ is the share of transport cost in the cost of imported inputs; $s_i^F = (1-\eta)^\xi\frac{p_{iF}^{1-\xi}}{p_x^{1-\xi}}$ is the share of imported inputs in the mix of intermediate inputs.

2. Gains from trade for importer:

$$\begin{aligned}\Omega_{ij} \equiv \pi_i - \pi_{i(-j)} &= (\mu_i-1)\mu_i^{-\sigma}P^\sigma Y\left(u_i^{1-\sigma} - \widetilde{u_i}^{1-\sigma}\right)\\ &= (\mu_i-1)u_iy_i\left(1-\left(1+s_i^F[(1+s_{i\tau}\Delta\tau)^{1-\xi}-1]\right)^{\frac{\beta(1-\sigma)}{1-\xi}}\right)\end{aligned}$$

3. Moreover:

$$\pi_i = (p_i - u_i)y_i = (\mu_i-1)\mu_i^{-\sigma}u_i^{1-\sigma}P^\sigma Y.$$
$$\frac{\partial\pi_i}{\partial\tau_{ij}} = (\mu_i-1)\mu_i^{-\sigma}(1-\sigma)u_i^{-\sigma}P^\sigma Y\frac{\partial u_i}{\partial\tau_{ij}}$$

[33] Without loss of generality, we abstract away from importers' idiosyncratic productivity.

$$
\begin{aligned}
&= (\mu_i - 1)\mu_i^{-\sigma}(1-\sigma)u_i^{-\sigma}P^\sigma Y \frac{\partial u_i}{\partial p_x}\frac{\partial p_x}{\partial p_{iF}}\frac{\partial p_{iF}}{\partial \tau_i}\frac{\partial \tau_i}{\partial \tau_{ij}} \\
&= (\mu_i - 1)\mu_i^{-\sigma}(1-\sigma)u_i^{-\sigma}P^\sigma Y \beta w^{1-\beta}p_x^{\beta-1}(1-\eta)^\xi \frac{p_{iF}^{-\xi}}{u_i^{-\xi}}\alpha_{it}\alpha_{ij}\frac{\tau_{ij}^{-\rho}}{\tau_i^{-\rho}} \\
&= (\mu_i - 1)p_i^{-\sigma}(1-\sigma)P^\sigma Y \beta w^{1-\beta}p_x^{\beta-1}\frac{q_{iF}}{x}\alpha_{it}\frac{t_{ij}}{t_i} \\
&= (\mu_i - 1)(1-\sigma)t_{ij}
\end{aligned}
$$

4. Determine bilateral prices:

$$
\max_{\tau_{ij}} \left(\pi_j - \pi_{j(-i)}\right)^{1-\phi}\left(\pi_i - \pi_{i(-j)}\right)^{\phi} \tag{26}
$$

The foc for the following problem is:

$$
\begin{aligned}
0 &= \frac{\partial \pi_j}{\partial \tau_{ij}} + \bar{\phi}\frac{\pi_j - \pi_{j(-i)}}{\pi_i - \pi_{i(-j)}}\frac{\partial \pi_i}{\partial \tau_{ij}} \\
&= 1 - \epsilon_{ij} + \epsilon_{ij}\frac{c_j}{\tau_{ij}} + \bar{\phi}\frac{(\tau_{ij} - c_j\widehat{\mu_{ij}})(\mu_i - 1)}{\Omega_{ij}}(1-\sigma)t_{ij} \\
&= -1 + \frac{\epsilon_{ij}}{\epsilon_{ij}-1}\frac{c_j}{\tau_{ij}} - \bar{\phi}\lambda_{ij} + \bar{\phi}\lambda_{ij}\frac{c_j}{\tau_{ij}}\widehat{\mu_{ij}} \\
\tau_{ij} &= c_j\left(\frac{1}{1+\bar{\phi}\lambda_{ij}}\bar{\bar{\mu}} + \frac{\bar{\phi}\lambda_{ij}}{1+\bar{\phi}\lambda_{ij}}\widehat{\mu_{ij}}\right)
\end{aligned}
$$

where $\lambda_{ij} = \frac{\sigma-1}{\epsilon_{ij}-1}\frac{(\mu_i-1)}{\Omega_{ij}}\frac{t_{ij}\tau_{ij}}{1}$

## A.3 Derivations of Quantitative Model

**Solution for the aggregate spending S** Let's decompose aggregate spending the following way

$$
\begin{aligned}
S &= S^C + S^{ROW} + S^X \\
&= I + \sum_i^N (1 - s_{iD})m_i + \sum_i^N s_{iD}m_i = I + \sum_i^N m_i
\end{aligned}
$$

Recall that

$$
\begin{aligned}
\pi_i &= (p_i - u_i)y_i = (p_i - \frac{\sigma-1}{\sigma}p_i)y_i \\
&= \frac{1}{\sigma}p_i y_i = \frac{1}{\sigma}p_i p_i^{-\sigma}P^\sigma Y\frac{P}{P} \\
&= \frac{1}{\sigma}\left(\frac{p_i}{P}\right)^{1-\sigma}PY = \frac{1}{\sigma}\left(\frac{p_i}{P}\right)^{1-\sigma}S \\
&\vdots
\end{aligned}
$$

$$\sum_{i=1}^{N} \pi_i = \sum_{i=1}^{N} \frac{1}{\sigma} \left(\frac{p_i}{P}\right)^{1-\sigma} S = \frac{1}{\sigma} S$$

so that we can write the representative consumer spending as

$$\begin{aligned} I &= L + \frac{1}{\sigma} S - \sum_{i=1}^{N} f \mathbf{1}(q_{iF} > 0)\} \\ &= L^{net} + \frac{1}{\sigma} S \end{aligned}$$

where $w = 1$ and $L^{net} = L - \sum_{i=1}^{N} f \mathbf{1}(q_{iF} > 0)\}$. Similarly, for the second element of the aggregate spending decomposition

$$\begin{aligned} \sum_{i=1}^{N} m_i &= \sum_{i=1}^{N} \beta \frac{\sigma - 1}{\sigma} p_i y_i \\ &= \sum_{i=1}^{N} \beta \frac{\sigma - 1}{\sigma} \left(\frac{p_i}{P}\right)^{1-\sigma} S \\ &= \beta \frac{\sigma - 1}{\sigma} S \end{aligned}$$

So that

$$\begin{aligned} S &= L^{net} + \frac{1}{\sigma} S + \beta \frac{\sigma - 1}{\sigma} S \\ &= L^{net} \frac{\sigma}{(1-\beta)(\sigma - 1)} \end{aligned}$$

## A.4 Derivations of Equations (17) and (18)

Let consider the consumers' aggregate price index: $P = \left[ \sum_{i=1}^{N} p_i^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$. We can compute the change in the price index as:

$$\begin{aligned} \log P &= \frac{1}{1-\sigma} \log\Big( \sum_{j=1}^{I} p_j^{1-\sigma} \Big) \\ \partial \log P &= \sum_i s_i \, \partial \log p_i, \qquad s_i \equiv \frac{p_i^{1-\sigma}}{\sum_j p_j^{1-\sigma}}. \end{aligned}$$

Given that $p_i = \mu \, u_i = \mu \frac{1}{\varphi_i} \, w^{1-\beta} \, p_{ix}^{\beta}$ and $p_{ix} = \left[ \eta^{\xi} \, P^{1-\xi} + (1-\eta)^{\xi} \left[ p_{iF} \, (1 + \bar{t}) + \tau_i \right]^{1-\xi} \right]^{\frac{1}{1-\xi}}$, we have:

$$\partial \log p_i = \beta \partial \log p_{ix} = \beta \left[ s_{iD} \partial \log P + (1 - s_{iD}) \left( (1 - s_{i\tau}) \partial \log \bar{t} + s_{i\tau} \partial \log \tau_i \right) \right]$$

Rearranging terms, we can derive the approximation in Equation (17).

We can derive the approximation for $\frac{\partial \log \tau_i}{\partial \log \bar{t}}$ in Equation (18) as follows:

$$\frac{\partial \log \tau_i}{\partial \log \bar{t}} \equiv \sum_j s_{ij} \frac{\partial \log \tau_{ij}}{\partial \log \bar{t}}$$
$$\frac{\partial \log \tau_{ij}}{\partial \log \bar{t}} \approx \frac{\partial \log \mu_{ij}}{\partial \log \bar{t}} + \frac{\partial \log c_j}{\partial \log \bar{t}} \equiv \frac{\partial \log \mu_{ij}}{\partial \log \tau_{ij}} \frac{\partial \log \tau_{ij}}{\partial \log \bar{t}} + \frac{\partial \log c_j}{\partial \log \tau_{ij}} \frac{\partial \log \tau_{ij}}{\partial \log \bar{t}} + \frac{\partial \log c_j}{\partial \log t_j} \frac{\partial \log t_j}{\partial \log \bar{t}},$$

where the first follows from $\tau_i = \left[ \sum_{j=1}^{J} \tau_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}$ and the second from $\tau_{ij} = \mu_{ij} c_j$.

Abstracting from general equilibrium effects (Alviarez et al., 2026), we can write:

$$\frac{\partial \log \tau_{ij}}{\partial \log \bar{t}} \approx = \Gamma_{ij} \frac{\partial \log \tau_{ij}}{\partial \log \bar{t}} + \Lambda_{ij} \frac{\partial \log \tau_{ij}}{\partial \log \bar{t}} + \frac{1-\theta}{\theta} \left( -\xi \sum_{i'} x_{i'j} (1 - s_{i'\tau} + s_{i'\tau} \frac{\partial \log \tau_{i'}}{\partial \log \bar{t}}) \right) \quad \text{where}$$
$$\Lambda_{ij} \equiv - \frac{\partial \log c_j}{\partial \log \tau_{ij}} = x_{ij} \frac{1-\theta}{\theta} \epsilon_{ij} \qquad \text{and}$$
$$\Gamma_{ij} \equiv - \frac{\partial \log \mu_{ij}}{\partial \log \tau_{ij}} = \omega_{ij} \frac{\widehat{\mu_{ij}}}{\mu_{ij}} \frac{\partial \log \widehat{\mu}_{ij}}{\partial \log \tau_{ij}} + (1-\omega_{ij}) \frac{\overline{\overline{\mu_{ij}}}}{\mu_{ij}} \frac{\partial \log \overline{\overline{\mu_{ij}}}}{\partial \log \tau_{ij}}$$
$$= \omega_{ij} \frac{\widehat{\mu_{ij}}}{\mu_{ij}} \left( \frac{x_{ij}(1-x_{ij})^{\frac{1}{\theta}-1}}{\theta - \theta(1-x_{ij})^{\frac{1}{\theta}}} - 1 \right) \rho(1-x_{ij}) + (1-\omega_{ij}) \frac{\overline{\overline{\mu_{ij}}}}{\mu_{ij}} \left( \frac{\rho - \epsilon_{ij}}{(\epsilon_{ij}-1)\epsilon_{ij}} (\rho-1)(1-s_{ij}) \right),$$

where in the first row we have used the fact that $t_j = \sum_{i \in I_j} t_{ij} = \sum_{i \in I_j} \frac{t_{ij}}{t_i} q_{iF} = \sum_{i \in I_j} x_{ij} q_{iF}$ and $\frac{\partial \log t_j}{\partial \log \bar{t}} = -\xi \sum_{i \in I_j} x_{ij} \frac{\partial \log (p_{iF}\,(1+\bar{t}))}{\partial \log \bar{t}}$.

Rearranging terms:

$$g_i = -\kappa \sum_{i'} \Big[ \sum_j s_{ij}\, \Phi_{ij}\, x_{i'j} \Big] \big(1 - s_{i'\tau} + s_{i'\tau}\, g_{i'}\big),$$

where $B_{ii'} = \sum_j s_{ij}\, \Phi_{ij}\, x_{i'j}$, $\kappa = \xi\, \frac{1-\theta}{\theta}$, and $\Phi_{ij} \equiv \frac{1}{1+\Gamma_{ij}+\Lambda_{ij}}$. Thus, using matrix notation we can write Equation (18):

$$\big[\mathbf{I} + \kappa\, \mathbf{B}\ \mathrm{diag}(\mathbf{s}_\tau)\big] \mathbf{g} = -\kappa\, \mathbf{B} \big(\mathbf{1} - \mathbf{s}_\tau\big),$$
$$\mathbf{g} = \Big[\mathbf{I} + \kappa\, \mathbf{B}\ \mathrm{diag}(\mathbf{s}_\tau)\Big]^{-1} \Big(-\kappa\, \mathbf{B} (\mathbf{1} - \mathbf{s}_\tau)\Big),$$

where $\mathbf{g} = (g_1, \ldots, g_I)^\top$ and $\mathbf{s}_\tau = (s_{1\tau}, \ldots, s_{I\tau})^\top$.

# B Customs Data

## B.1 Cleaning

We perform some standard cleaning of the transactions reported in the custom to reduce the noise coming from possible mistakes and misreporting in the declarations. Our initial dataset comprises around 30 million transactions. We drop those for which the CIF reported is lower than the FOB value and those for which there was a discrepancy between FOB + reported additional costs and CIF larger than 10%. We also drop all those transactions that are missing the country of origin or that are missing other key information. We also drop sectors such as Arms and Ammunition (HS 93) and Antiques and Art (HS 97).

For the transportation sector, we combine air transport and couriers into a single category, and then we keep only those transactions that report as mode of transport sea, air, or road (99.99% of the sample). We also drop all the transactions that are reported as via land but are with countries that are either implausibly far or not connected at all via land. After this preliminary cleaning we have a dataset of more than 28 million transactions of Chilean firms importing from the rest of the world.

### B.1.1 Multi-product transaction

One issue with the transaction-level data is that for some multi-product transactions, we cannot observe the total freight cost for each product, which we will need to use to build our transportation cost variable. To overcome this problem, we build the share of each product, within the transaction, in terms of quantity and assign the weight accordingly.

### B.1.2 Shippers cleaning

The final step is to clean the shippers' names in order to have a precise idea of the firm in charge of the transportation of the goods in Chile. To do so, we use two key pieces of information from the custom declaration. First, we observe a string variable reporting the name of the shipper which we clean manually. Second, we have information on the shippers' RUT (Rol Único Tributario) which is a unique tax code that each company has for tax purposes in Chile. We then match this RUT to a list of foreign transporters provided by the Chilean Government in order to further clean and homogenize the list of transportation firms. As the final step, we replace companies' names with the parent owner to reduce the number of firms with the same ownership in the same market. For example, we replace the company name with Lufthansa when we have Swissair or CSCL with COSCO after its acquisition in 2015. Table B.1 provides an example of the cleaned names for the top companies, in terms of value shipped in 2019, in our sample.

Table B.1: Cleaned Names and Top Companies' Share

| Sea | | Air | | Road | |
|---|---|---|---|---|---|
| Transport Company | Market Share | Transport Company | Market Share | Transport Company | Market Share |
| MAERSK | 19.40 | LAN CARGO | 44.68 | PASTENES GUTIERREZ CATALINA ROCIO | 8.68 |
| HAPAG LLOYD | 16.35 | ATALS AIR - POLAR AIR | 15.67 | MEDINA ENRIQUEZ JUSTO EDUARDO | 5.13 |
| ONE | 8.42 | AIR FRANCE | 11.63 | ABC CARGAS | 4.89 |
| MSC | 7.75 | IBERIA | 5.40 | ORTEGA CASANOVA MANUEL ERNESTO | 4.66 |
| CMA-CMG | 5.68 | AVIANCA | 4.97 | BECERRA VALENZUELA OCTAVIO MIGUEL | 4.62 |

**Notes**: The Table reports the top 5 companies, in terms of value shipped in the year 2019 across all product, used by Chilean companies to import goods. Source: 2019 Chilean Custom.

# C The Incoterms rules

The Incoterms, International Commerce Terms, are a set of standards used in international and domestic contracts for the delivery of goods and are established by the International Chamber of Commerce (ICC) (International Chamber of Commerce, 2021). These rules define the delivery terms for each transaction. All transactions can be divided into two main groups depending on whether it is the importer's or the exporter's responsibility to arrange the international shipping of the goods. In particular, each transaction can be ranked in terms of the importer's responsibility in the delivery process. For instance, the importer plays a fully passive role in the case the agreed term is the so-called DDP (Delivered Duty Paid) which places the greatest burden on the exporter. In this case, the exporter agrees to clear the goods through customs at the destination and also to deliver the goods at a previously specified location. By contrast, under the EXW (Exworks-Factory) the seller has the minimum obligations. Indeed, it is the importer's responsibility to move the goods from a designated factory of production to the desired final location. Following standard classification, we group transactions that fall under the category of EXW, FCA (Free Carrier), FOB (Free on Board), and FAS (Free alongside ship) as transactions in which it is the importer's responsibility to arrange the international shipping of the goods. By contrast, transactions falling into the terms of CFR (Cost and Freight), CIF (Cost, Insurance and Freight), DDP (Delivered Duty Paid), CPT (Carriage Paid to), DAP (Delivered at Place) are characterized by the fact that it is the seller's responsibility to negotiate and pay for the shipping of the goods. Table C.1 reports all cases and the duties assigned to the importer and the exporter.

Figure C.1 reports the share of transaction, and value, by mode for each Incoterms category. We can see that for Rail, and partially for air, we have many observations for which the shipment arrangements are not reported. Of the remaining observations, we can see that the predominant delivery terms are ones in which the burden of the transport is on the importer. This is especially true when we look at ocean shipping in the left panel.

Table C.1: Allocation of obligations and charges under Incoterms 2020

| | Departure | | | | C-terms | | | | Arrival | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Obligation / Charge | EXW | FCA | FAS | FOB | CFR | CIF | CPT | CIP | DAP | DPU | DDP |
| Export packaging | S | S | S | S | S | S | S | S | S | S | S |
| Loading charges | B | S | S | S | S | S | S | S | S | S | S |
| Delivery to port/place | B | S | S | S | S | S | S | S | S | S | S |
| Export duty, taxes & customs clearance | B | B | S | S | S | S | S | S | S | S | S |
| Origin terminal charges | B | B | S | S | S | S | S | S | S | S | S |
| Loading on carriage | B | B | B | S | S | S | S | S | S | S | S |
| Carriage charges | B | B | B | B | S | S | S | S | S | S | S |
| Insurance | N | N | N | N | S | S | N | N | N | N | N |
| Destination terminal charges | B | B | B | B | B | B | B | B | S | S | S |
| Delivery to destination | B | B | B | B | B | B | B | B | S | S | S |
| Unloading at destination | B | B | B | B | B | B | B | B | B | S | S |
| Import duty, taxes & customs clearance | B | B | B | B | B | B | B | B | B | B | S |

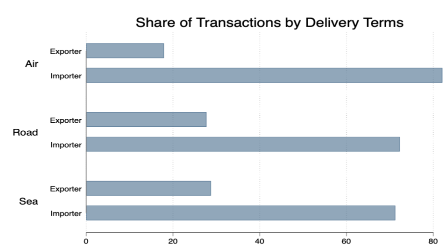
**Notes**: S=Seller's responsibility; B=Buyer's responsibility; N=Negotiable between parties. Columns are grouped by risk-transfer point: Departure terms (EXW–FOB), C-terms where the seller pays carriage but risk transfers at origin (CFR–CIP), and Arrival terms where risk transfers at destination (DAP–DDP). Source ICC. .

Table C.2: Statistics by Incoterm

| Incoterm | Mode | Mean Unit Freight | Mean FOB | Median Unit Freight | Median FOB |
|---|---|---|---|---|---|
| Exporter | Air | 54.46 | 20359.40 | 8.83 | 1741.83 |
| Importer | Air | 83.95 | 10170.07 | 8.08 | 1700.00 |
| Unknown | Air | 105.49 | 19656.27 | 7.97 | 1373.57 |
| Exporter | Road | 1.40 | 136534.22 | 0.18 | 27479.06 |
| Importer | Road | 4.47 | 50610.27 | 0.46 | 6955.71 |
| Unknown | Road | 16.12 | 105601.40 | 0.26 | 15000.00 |
| Exporter | Sea | 5.99 | 116745.77 | 0.22 | 12217.13 |
| Importer | Sea | 5.59 | 45750.04 | 0.39 | 4828.84 |
| Unknown | Sea | 6.68 | 179780.81 | 0.40 | 3903.67 |

**Notes**: Summary statistics by Incoterm, which represents the party in charge of arranging and paying for the delivery of the imported good. Source Chilean Custom.

Figure C.1: Party Arranging Import Transactions




**Notes**: The Figure reports the share of transactions for which we observe Incoterms that are arranged by the importer across different transport modes. The party in charge of the transaction is reported in the variable Incoterms included in the Chilean custom data.

# D Manufacturing Data - ENIA

We use data from the Annual Survey of Manufacturing (ENIA), administrated by the Chilean National Institute of Statistics (INE), covering the years 1995 - 2019. The data is at the establishment-year level and includes approximately 30 manufacturing industries, with roughly 4,000 observations per year. For each observation, the dataset contains information on capital stock, value added, labor, wage bill, domestic and imported materials, revenues, and electricity consumption.[34] Capital stock data are unavailable after 2015. We extend the sample to 2019 by constructing the capital stock using investment and depreciation data via a perpetual inventory method. Industries are defined at the three-digit CIIU Rev. 3 level (Chilean industry classification).

Observations with zero or negative values for capital, materials, revenues, electricity consumption, or wage bill are excluded. Additionally, we drop observations with a labor share or materials share of revenue exceeding one. To remove outliers, we exclude observations in the bottom and top 5% of labor and materials shares of revenue for each industry.

A firm-level measure of capital costs is constructed as the product of capital stock and the rental rate net of depreciation. The average real interest rate for Chile during the sample period, reported in the World Bank World Development Indicators, serves as a proxy for the rental rate of capital (Raval, 2023).[35] The rental rate is combined with sectoral depreciation rates from Oberfield and Raval (2021), after creating a concordance between NAICS and CIIU classifications.

**Calibration and Moments Details** Under the assumption of constant return to scale, as in our theoretical framework, Autor et al. (2020) shows that markup can be measured as the ratio of firm sales to total costs:

$$\mu_{it} = \frac{\alpha_{it}^v}{S_{it}^v} = \frac{\texttt{Sales}_{it}}{\texttt{Total Cost}_{it}}, \tag{27}$$

where $\alpha_{it}^v$ and $S_{it}^v$ represent the output elasticity and the factor share of input $v$ ($S_{it}^v = \frac{\texttt{Expenditure on } v}{\texttt{Sales}_{it}}$), respectively. The second equality follows from the CRS assumption, i.e. $\alpha_{it}^v = \frac{\texttt{Expenditure on } v}{\texttt{Total Cost}_{it}}$. In mapping Equation (27) to the data, we assume that total costs are equal to the sum of wage bill, materials expenditure, electricity expenditure, and capital costs. We calibrate $\sigma$ to match the median markup in the economy, which delivers back a value of 6.

Given the implied $\sigma$, we calibrate the share of material in the production, $\beta$, leveraging the observed factor shares, $\beta \equiv \frac{m_i}{p_i y_i} = \beta \frac{\sigma - 1}{\sigma}$. In mapping it to the data, we define materials as the sum of intermediate inputs and electricity consumption.

We estimate the production function in value added applying Levinsohn and Petrin (2003)

[34] The INE applies a small amount of noise to all variables to ensure statistical privacy. For integer variables, such as labor, we use the floor of the value reported by INE.

[35] The real interest rate represents the private sector lending rate, adjusted for the domestic inflation rate as measured by the GDP deflator.

and Ackerberg et al. (2015) techniques, using labor as a variable input and electricity consumption as a proxy variable. We specify the production function as follows

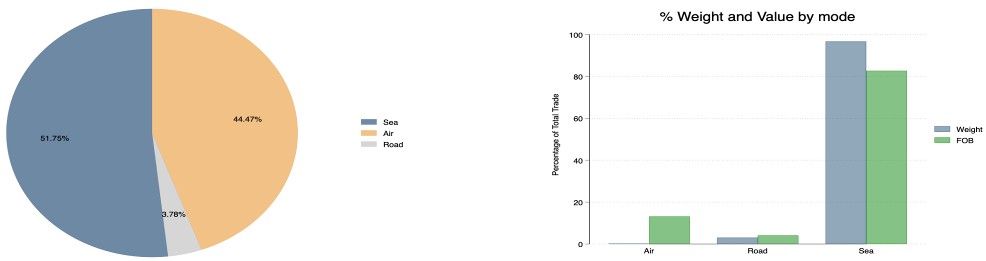
$$y_i = A\varphi_i l_i^{1-\beta} m_i^{\beta} s_{iD}^{-\frac{\beta}{\gamma-1}},$$

in order to estimate $\gamma$ using the variation in domestic expenditure shares $s_{iD}$. The share is defined in terms of domestic and imported intermediate inputs. We drop observations with a negative domestic share and trim values at the 5% level within each industry.

Lastly, we leverage ENIA to compute moments useful for the estimation of the moment. We compute the aggregate domestic share as the value-added weighted share of domestic input across firms in the sample. We compute the within-industry dispersion of sales share using firms' revenues, and use the average across industries as targeted moments.

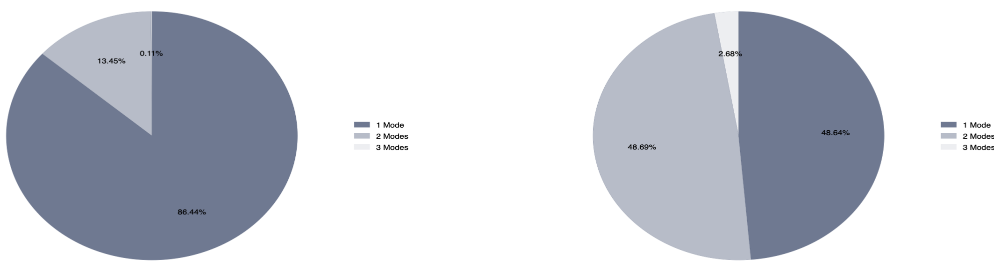
# E Additional Stylized Facts

Figure E.1: Trade Volumes and Number of Transaction by Mode



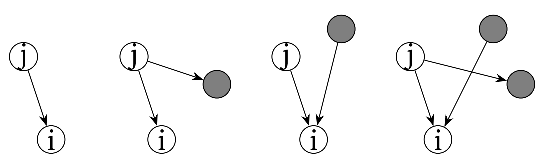

**Notes**: The left panel reports the share of total transactions by transport mode. The right panel reports the corresponding shares of total trade value (navy) and weight (green). Rail transport is recorded in the customs data but accounts for less than 1% of total trade and is excluded from the sample.

Figure E.2: Number of Modes




**Notes**: This figure reports the share of importers using one or more transport modes. The left panel conditions on importer-origin-product (HS6) cells; the right panel conditions on importer-origin cells.

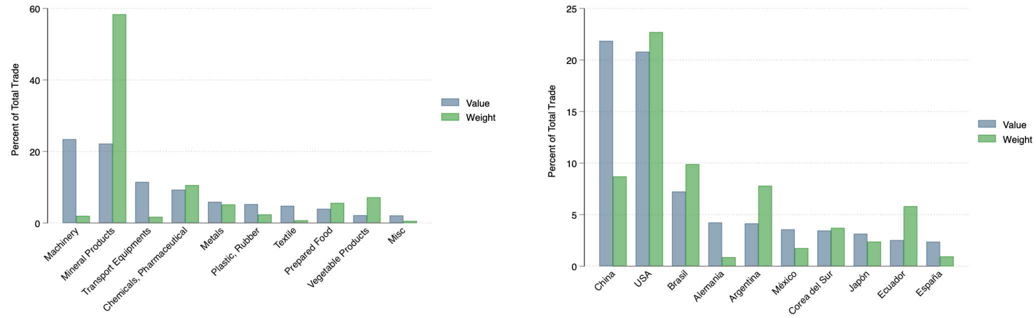
Figure E.3: Network Structure in International Shipping




| | 1:1 | 1:m | m:1 | m:m |
|---|---|---|---|---|
| Fraction of links | 0.216 | 0.381 | 0.041 | 0.362 |
| | (0.355) | (0.284) | (0.133) | (0.322) |
| Fraction of import value | 0.230 | 0.343 | 0.055 | 0.372 |
| | (0.375) | (0.310) | (0.160) | (0.351) |

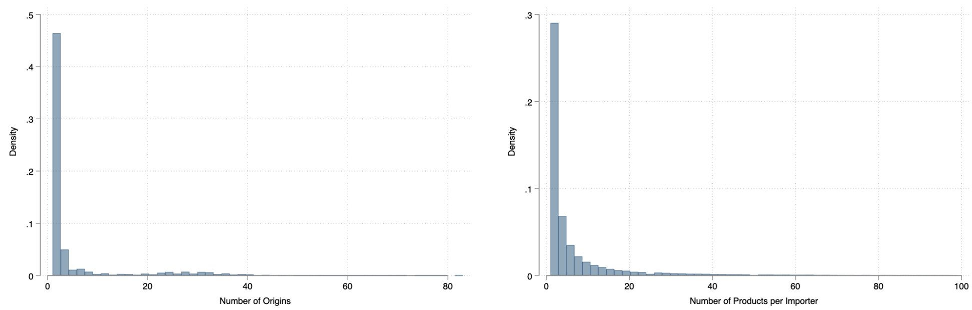
**Notes**: This figure illustrates the network structure of carrier-importer relationships in the sample. Each cell reports the median share across origin $\times$ transport mode markets, with standard deviations in parentheses. Links are measured both by number of matches (top row) and by trade value (bottom row). $j$ denotes the carrier, $i$ denotes the importer.

Figure E.4: Import Composition by Sector and Origin



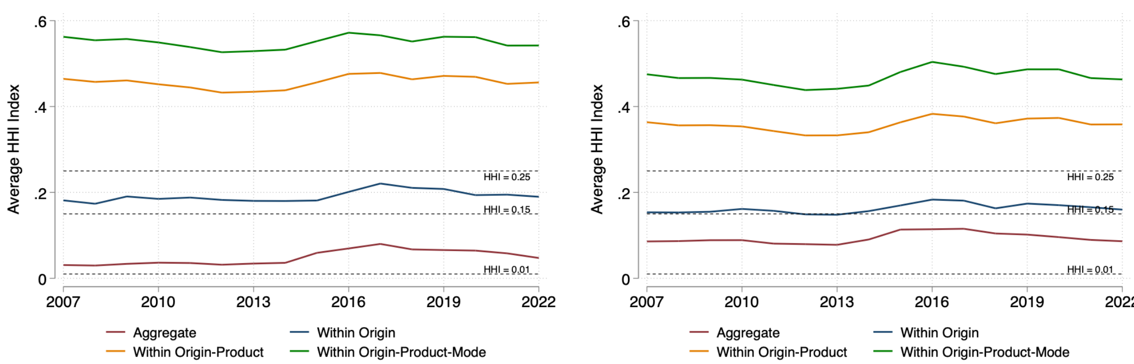

**Notes**: This figure decomposes Chilean imports by sector (left panel) and country of origin (right panel). A sector is defined as a HS2 category. In both figures, the bars are in descending order based on their total value of trade.

Figure E.5: Numbers of Origins and Product by Importer



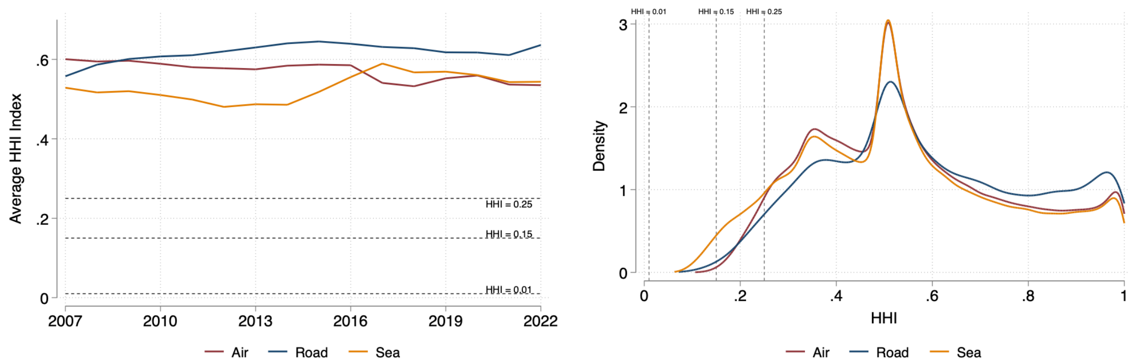

**Notes**: The left panel reports the distribution of origins per importer. The right panel reports the distribution of products (HS6) per importer.

Figure E.6: Concentration in International Transportation



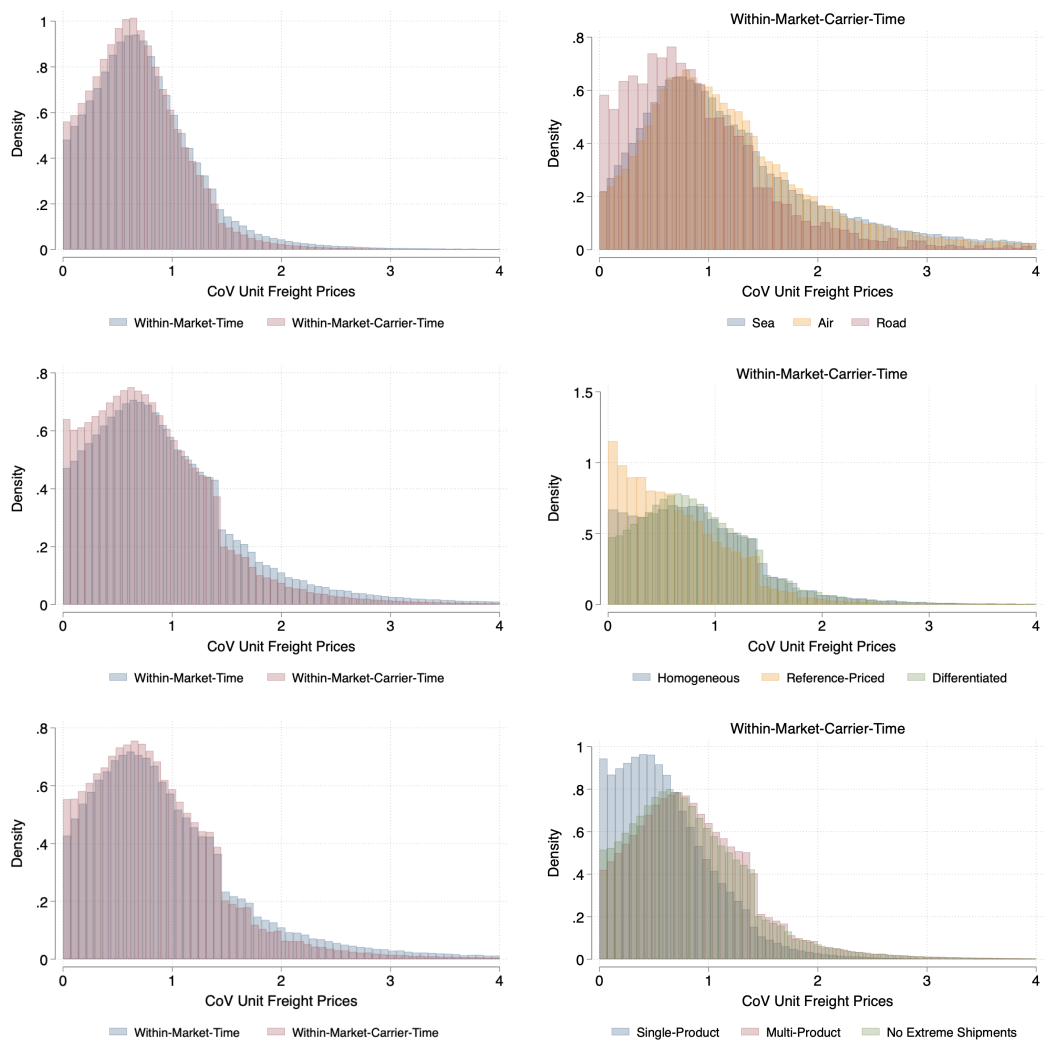

**Notes**: The right panel plots the average HHI index across the different markets of the transportation sector over time. Markets are defined according different levels of granularity. The red line considers a unique aggregate transportation market. The blue line defines markets by the origin country. The orange and green lines defines markets as a combination of origin-product and origin-product-mode, respectively. A product is defined as a HS6 category. Modes are sea, air, and road freight. Carriers' market share are computed in terms of value (FOB in USD) in the left panel and number of transactions in the right panel.

Figure E.7: Concentration in International Transportation by Mode




**Notes**: The left panel plot the average HHI index across the different markets of the transportation sector over time. Markets are defined as a mode-origin-product combination, where a product is defined as a HS6 category. We compute the average distinguishing markets by their mode (sea, air, and road freight). The left panel plots the distribution of HHI indices across the different markets, distinguishing markets by their mode (sea, air, and road freight). Carriers' market share are computed in terms of value shipped.

Figure E.8: Unit Freight Price Dispersion




**Notes**: Markets in all panels are defined as a origin-mode-product triplet, where modes are sea, air, and road, and products are HS6 categories, respectively. Time is measured at an annual frequency. The top left panel plots the distribution of the coefficient of variation of unit freight prices computed in terms of value (FOB in USD). In the top right panel, within-market-carrier-time CoVs are reported separately across modes. The middle left panel plots the distribution of the coefficient of variation of unit freight prices within a market-time and within market-carrier-time using the full sample of transactions. The middle right panel plots the distribution of the coefficient of variation of unit freight prices within market-carrier-time distinguishing products according to the Rauch (1999) classification. The bottom left panel plots the distribution of the coefficient of variation of unit freight prices within a market-time and within market-carrier-time where prices are expressed in cost per TEU, using product-level kg-to-TEU conversion coefficients. The bottom right panel plots the within-market-carrier-time distribution separately for single-product, multi-product transactions, and the full sample excluding the bottom and top five percent of the shipment size distribution measured by FOB value.

Table E.1: Fixed-effect Decomposition of Freight Price Dispersion - Alternative Measures

| | Value | | Quantities | | Homogeneous | | TEU | | Full Sample | | Single-Product | | Multi-Product | | No Extreme Shipments | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) | (14) | (15) | (16) |
| **Panel A - Share of price dispersion explained by:** | | | | | | | | | | | | | | | | |
| Observables | . | 0.032 | . | -0.001 | . | 0.007 | . | 0.001 | . | 0.013 | . | 0.108 | . | -0.002 | . | 0.007 |
| Buyer FE | 0.114 | 0.106 | 0.081 | 0.080 | 0.084 | 0.084 | 0.092 | 0.092 | 0.035 | 0.036 | 0.043 | 0.036 | 0.060 | 0.060 | 0.050 | 0.051 |
| Transport Company FE | 0.022 | 0.022 | 0.025 | 0.025 | 0.079 | 0.078 | 0.003 | 0.003 | 0.040 | 0.041 | 0.044 | 0.038 | 0.023 | 0.023 | 0.034 | 0.034 |
| Product x Time x Origin x Mode | 0.453 | 0.440 | 0.684 | 0.686 | 0.613 | 0.607 | 0.395 | 0.395 | 0.664 | 0.652 | 0.793 | 0.709 | 0.619 | 0.622 | 0.653 | 0.646 |
| Match Residual | 0.412 | 0.400 | 0.210 | 0.210 | 0.224 | 0.224 | 0.510 | 0.510 | 0.260 | 0.259 | 0.121 | 0.108 | 0.298 | 0.298 | 0.263 | 0.262 |
| **Panel B - Within Carrier-Product-Origin-Time-Mode:** | | | | | | | | | | | | | | | | |
| Observables | . | 0.049 | . | 0.001 | . | 0.003 | . | 0.001 | . | 0.009 | . | 0.219 | . | 0.001 | . | 0.006 |
| Buyer FE | 0.232 | 0.208 | 0.171 | 0.170 | 0.260 | 0.259 | 0.150 | 0.150 | 0.095 | 0.094 | 0.174 | 0.139 | 0.143 | 0.143 | 0.122 | 0.122 |
| Match Residual | 0.768 | 0.743 | 0.829 | 0.830 | 0.740 | 0.738 | 0.850 | 0.849 | 0.905 | 0.897 | 0.826 | 0.642 | 0.857 | 0.857 | 0.878 | 0.872 |

**Notes**: The table reports the results of a statistical decomposition exercise based on OLS regressions on the estimating specification in Equation (1). Unit freight prices are computed by dividing total freight cost by value (FOB), columns (1)-(2), or by quantities, columns (3)-(4). In columns (5)-(6) we restrict the sample to homogeneous goods following the Rauch (1999) classification. In columns (7)-(8) unit freight prices are expressed in cost per TEU, using product-level kg-to-TEU conversion coefficients. In columns (9)-(10) we do not restrict the analysis to transactions arranged by importers (full sample). Columns (11)-(12) and columns (13)-(14) restrict the sample to single-product transactions and to multi-product transactions, respectively. Columns (15)-(16) exclude the bottom and top five percent of the shipment size distribution, measured by FOB value at the importer-carrier-origin-mode-product-quarter level. For each specification, the even column includes observable characteristics (carrier experience, age of importer-carrier relationship, and transaction size), while the odd column includes fixed effects only. A product is defined as a HS6 category and time is measured at quarterly frequency.

Table E.2: Prices and Bilateral Concentration - Robustness

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| | Sea | Air | Quantity | Homogeneous | TEU | Full Sample | Single-Product | Multi-Product | No Extreme Shipments |
| Log Carrier Share | 0.373 | 0.445 | 0.169 | 0.389 | 0.421 | 0.381 | 0.326 | 0.565 | 0.460 |
| | (0.076) | (0.052) | (0.064) | (0.281) | (0.070) | (0.027) | (0.079) | (0.046) | (0.056) |
| Log Importer Share | -0.420 | -0.555 | -0.246 | -0.467 | -0.466 | -0.471 | -0.391 | -0.610 | -0.555 |
| | (0.069) | (0.048) | (0.061) | (0.295) | (0.064) | (0.025) | (0.071) | (0.043) | (0.045) |
| $N$ | 631912 | 265141 | 904380 | 22712 | 566355 | 1776916 | 214122 | 501011 | 675306 |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| $FE_{jmt} + FE_{imt}$ | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| KP F-stat | 59.006 | 85.879 | 123.458 | 11.984 | 65.429 | 245.835 | 32.360 | 94.535 | 90.308 |
| Hansen J | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |

**Notes**: The table reports the estimates from the specification in Equation (2) estimated using IV. Column (1) and column (2) restrict the sample to sea and air freight, respectively. Column (3) measures unit freight prices per quantity shipped rather than per kilogram. Column (4) restricts the sample to homogeneous goods following the Rauch (1999) classification. Column (5) expresses unit freight prices in cost per TEU, using product-level kg-to-TEU conversion coefficients. Column (6) uses the full sample of transactions, without restricting to shipments arranged by the importer. Columns (7) and (8) restrict the sample to single-product transactions and to multi-product transactions, respectively. Column (9) excludes the bottom and top five percent of the shipment size distribution, measured by FOB value at the importer-carrier-origin-mode-product-quarter level. All columns include the age of the importer–carrier relationship as an additional control, and carrier-market-time and importer-market-time fixed effects. Note that in column (1) and (2) the market fixed effects do not include a mode dimension. Markets in all panels are defined as a origin-mode-product triplet, products are HS6 categories, and time is measured at quarterly frequency. Standard errors are clustered at the importer level. We exclude all importer-market-time and carrier-market-time singletons from the estimation.

Table E.3: Prices and Bilateral Concentration - Relationship Tenure Interaction

| | (1) |
|---|---|
| | Full sample |
| Log Carrier Share | 0.428 |
| | (0.052) |
| Log Importer Share | -0.519 |
| | (0.043) |
| Log Carrier Share $\times$ Tenure | 0.001 |
| | (0.002) |
| Log Importer Share $\times$ Tenure | 0.000 |
| | (0.001) |
| Controls | Yes |
| $FE_{jmt} + FE_{imt}$ | Yes |
| KP F-stat | 63.189 |
| p-val: both interactions = 0 | 0.567 |
| $N$ | 898329 |

**Notes**: The table estimates the IV specification in Equation (2) on the full sample, augmented with interactions between each bilateral concentration measure and relationship tenure. The two concentration measures and their two tenure interactions are treated as endogenous and instrumented with the two baseline instruments and their interactions with tenure. The row "p-val: both interactions = 0" reports the $p$-value of the joint test that both interaction coefficients are zero. Markets are defined as an origin-mode-product triplet, products are HS6 categories, and time is measured at quarterly frequency. Standard errors are clustered at the importer level. We exclude all importer-market-time and carrier-market-time singletons from the estimation.

# F Additional Empirical Results

Table F.1: Estimated $\widehat{\rho}$ - Robustness FEs

| | (1) OLS | (2) IV | (3) IV | (4) IV | (5) IV |
|---|---|---|---|---|---|
| $\widehat{\beta}$ | 0.050 | -1.644 | -2.729 | -2.548 | -1.783 |
| | (0.019) | (0.519) | (0.574) | (0.580) | (0.588) |
| $N$ | 149761 | 149364 | 149364 | 145652 | 145652 |
| Implied $\widehat{\rho}$ | . | 2.644 | 3.729 | 3.548 | 2.783 |
| | | $FE_j + FE_m$ | $FE_j + FE_m + FE_t$ | $FE_j \times FE_m + FE_t$ | $FE_j \times FE_m$ |
| KP F-stat | | 17.12 | 15.37 | 14.54 | 14.50 |
| Hansen J p-value | | 0.0674 | 0.641 | 0.801 | 0.322 |

**Notes**: The table reports the estimated price elasticities. Column (1) is estimated via OLS without any fixed effect. Columns (2) to (4) saturate the specification in difference with different sets of fixed effects, using the set of instruments from the main specification. All specifications are estimated in difference. Standard errors are clustered at the importer level. Implied $\widehat{\rho}$ reports the implied $\rho$, computed as $\widehat{\rho} = -\widehat{\beta} + 1$.

Table F.2: Estimated $\widehat{\rho}$ - Robustness IVs

| | (1) Hausman | (2) BLP | (3) Hausman + $N_j$ | (4) Hausman + $N_i$ | (5) Full |
|---|---|---|---|---|---|
| $\widehat{\beta}$ | -2.137 | -2.587 | -2.492 | -1.843 | -2.548 |
| | (3.174) | (0.535) | (0.565) | (1.191) | (0.580) |
| Implied $\widehat{\rho}$ | 3.137 | 3.587 | 3.492 | 2.843 | 3.548 |
| KP rk Wald F | 1.33 | 20.06 | 21.67 | 3.51 | 14.54 |
| Hansen J p-value | . | 0.602 | 0.905 | 0.895 | 0.801 |
| IV family | Hausman | $N_i$ + $N_j$ | Hausman + $N_j$ | Hausman + $N_i$ | Full |
| N | 145652 | 145748 | 145652 | 145652 | 145652 |

**Notes**: The table reports the estimated price elasticities using different combinations of the IVs used in the main specification. Column (1) employs on the Hausman-type IV; Columns (2) uses only the (log) numbers of agents in the market; Column (3) and (4) use the Hausman-type IV with the number of carriers and the number of importers, respectively. All specifications are estimated in difference. Standard errors are clustered at the importer level. Implied $\widehat{\rho}$ reports the implied $\rho$, computed as $\widehat{\rho} = -\widehat{\beta} + 1$.

Table F.3: Estimated $\widehat{\rho}$ - Robustness Alternative Definition

| | (1) Second-Lowest | (2) Alt. Trim | (3) Alt. Market | (4) TEU | (5) Single-Product | (6) Multi-Product | (7) No Extreme Shipments |
|---|---|---|---|---|---|---|---|
| $\widehat{\beta}$ | -3.508 | -1.768 | -3.241 | -2.794 | -3.274 | -3.135 | -1.845 |
| | (0.509) | (0.373) | (1.404) | (0.772) | (0.886) | (0.927) | (0.403) |
| $N$ | 145632 | 175814 | 211578 | 91739 | 37012 | 78162 | 119826 |
| Implied $\widehat{\rho}$ | 4.508 | 2.768 | 4.241 | 3.794 | 4.274 | 4.135 | 2.845 |
| $FE_j \times FE_m + FE_t$ | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| KP F-stat | 25.83 | 21.07 | 3.755 | 10.24 | 7.081 | 5.954 | 19.77 |
| Hansen J p-value | 0.798 | 0.223 | 0.161 | 0.332 | 0.830 | 0.474 | 0.172 |

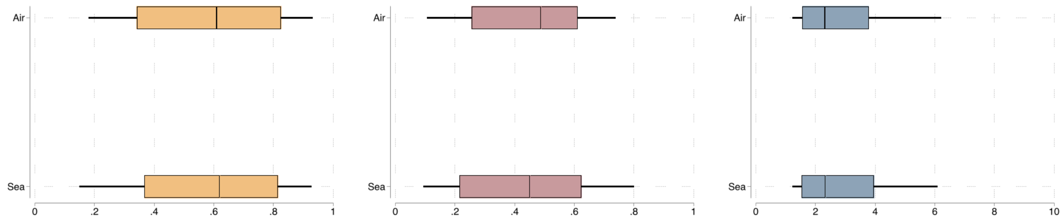
**Notes**: The table reports the estimated price elasticities. Column (1) uses the second-lower carrier when taking the first difference in Equation (8). In Columns (2), we use a less restrictive cleaning in selecting the dataset (trimming unit freight prices at 2.5 percent rather than 5 percent). Column (3) defines the market of relevance for carriers and importers, and thus the variables, at the origin-mode level, rather than origin-mode-product. In this case we use the number of importers and its log transformation as IVs. Column (4) uses prices and bilateral shares defined in terms of TEU, rather than weight shipped. Columns (5) and (6) restrict the sample to single-product transactions and to multi-product transactions, respectively. Column (7) excludes the bottom and top five percent of the shipment size distribution, measured by FOB value at the importer-carrier-origin-mode-product-quarter level. All specifications are estimated in difference. Standard errors are clustered at the importer level. Implied $\widehat{\rho}$ reports the implied $\rho$, computed as $\widehat{\rho} = -\widehat{\beta} + 1$.

Table F.4: Estimated $\widehat{\overline{\phi}}$ and $\widehat{\theta}$ - Robustness Alternative Definition

| | (1) Alt. Market | (2) Alt. Trim | (3) TEU | (4) Single-Product | (5) Multi-Product | (6) No Extreme Shipments |
|---|---|---|---|---|---|---|
| $\widehat{\overline{\phi}}$ | 2.893 | 3.834 | 8.348 | 2.385 | 4.987 | 4.158 |
| | (0.672) | (1.960) | (14.393) | (1.467) | (4.343) | (2.671) |
| Implied $\widehat{\phi}$ | 0.743 | 0.793 | 0.893 | 0.705 | 0.833 | 0.806 |
| $\widehat{\theta}$ | 0.482 | 0.513 | 0.568 | 0.385 | 0.562 | 0.494 |
| | (0.215) | (0.041) | (0.067) | (0.072) | (0.059) | (0.050) |
| KP $F$-stat | 143.936 | 2241.684 | 1408.444 | 1443.536 | 1021.448 | 1771.635 |
| Hansen $J$-stat | 0.154 | 1.835 | 2.011 | 2.700 | 1.459 | 1.218 |
| Hansen $J$ $p$-value | 0.926 | 0.399 | 0.366 | 0.259 | 0.482 | 0.544 |
| $N$ | 6894456 | 1002601 | 486114 | 161671 | 355317 | 640959 |

**Notes**: The table reports the estimated bargaining weight $\widehat{\overline{\phi}}$ and scale parameter $\widehat{\theta}$, with the baseline specification from Table 4 as the point of comparison. Column (1) defines markets at the origin-mode level rather than origin-mode-HS6-product; instruments are the mean and median buyer and seller bilateral shares and their non-linear transformations, computed in the market excluding the pairs involved in each moment; Column (2) trims unit freight prices at the 2.5th percentile within each market rather than the 5th. Column (3) measures prices and bilateral shares in terms of TEUs rather than kilograms. Columns (4) and (5) restrict the sample to single-product transactions and to multi-product transactions, respectively. Column (6) excludes the bottom and top five percent of the shipment size distribution, measured by FOB value at the importer-carrier-origin-mode-product-quarter level. Instruments in Columns (2)-(6) are the mean and median buyer and seller bilateral shares, excluding the pairs involved in each moment; observations with $|\log \tau_{ijmt} - \log \tau_{kjmt}| > 0.5$ are dropped. All specifications are estimated in differences. Standard errors are robust. Implied $\widehat{\phi}$ is recovered from $\widehat{\overline{\phi}}$ via $\phi = \overline{\phi}/(1 + \overline{\phi})$.

Figure F.1: Distribution Parameters by Mode of Transportation



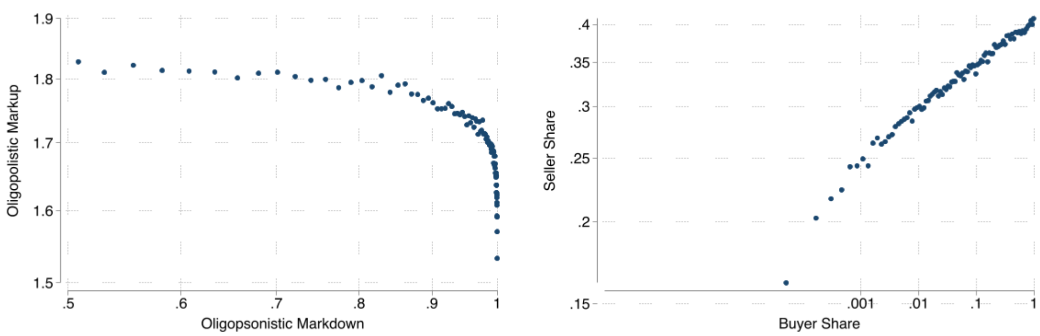

**Notes**: The figure plots the distribution of $\widehat{\phi}$ (left), $\widehat{\theta}$ (center), and $\widehat{\rho}$ (right) estimated at the origin-HS6-product-mode level, separately for sea and air shipments (road dropped for limited number of observations). Each box spans the interquartile range; whiskers extend from the 10th to the 90th percentile. Values of $\widehat{\rho} > 10$ are excluded. Parameters are estimated as described in Table 5.

Table F.5: Correlation Estimated Parameters - Market Characteristics

| | Number of Agents | | | Concentration | | |
|---|---|---|---|---|---|---|
| | $\widehat{\phi}$ | $\widehat{\theta}$ | $\log \widehat{\rho}$ | $\widehat{\phi}$ | $\widehat{\theta}$ | $\log \widehat{\rho}$ |
| Number of Importers | -0.035 | 0.039 | -0.112 | | | |
| | (0.014) | (0.011) | (0.034) | | | |
| Number of Carriers | 0.002 | -0.018 | 0.122 | | | |
| | (0.028) | (0.009) | (0.060) | | | |
| HHI(s) | | | | 0.015 | 0.013 | 0.023 |
| | | | | (0.014) | (0.010) | (0.046) |
| HHI(x) | | | | 0.035 | -0.016 | 0.071 |
| | | | | (0.013) | (0.011) | (0.042) |
| $N$ | 1551 | 1551 | 1146 | 1551 | 1551 | 1146 |

**Notes**: The table reports OLS estimates from regressions of $\widehat{\phi}$, $\widehat{\theta}$, and $\widehat{\rho}$ on market-level characteristics, estimated at the origin-HS6-product-mode level. Columns (1)-(3) use the average number of importers and carriers in the market as regressors. Columns (4)-(6) replace competitor counts with the average HHI indices of the bilateral shares $s_{ijmt}$ and $x_{ijmt}$, constructed at the origin-HS6-product-mode-time level and averaged over time. All specifications absorb transport mode fixed effects and weight observations by the number of importer-carrier pairs per market. Standard errors are clustered at the HS2 sector level. Markets with $\widehat{\rho} > 50$ are excluded.

Figure F.2: Correlation Markups and Markdown




**Notes**: The left panel plots the binscatter relationship between the log oligopolistic markup $\overline{\overline{\mu}}_{ijmt}$ and the log oligopsonistic markdown $\widehat{\mu}_{ijmt}$, after absorbing origin-HS6-product-mode-time fixed effects. Values of $\overline{\overline{\mu}}_{ijmt} > 3$ are trimmed. The right panel plots the binscatter relationship between the log bilateral shares $s_{ijmt}$ and $x_{ijmt}$, after absorbing origin-HS6-product-mode-time fixed effects. $s_{ijmt}$ is the share of carrier $j$ in importer $i$'s total transportation costs in market $m$ at time $t$; $x_{ijmt}$ is the share of importer $i$ in carrier $j$'s total shipments in market $m$ at time $t$. Markups are constructed using the estimated parameters from Table 4.

# G Converting Weight to TEU

We use proprietary Chilean customs data from Panjiva. The dataset reports the same information as the publicly accessible database, with some additional internally-generated variables. Importantly, the dataset reports the shipment volume expressed in twenty-foot equivalent units, `volumeteu`.

However, Panjiva does not disclose the procedure generating `volumeteu`, so we characterize the mapping empirically. The variable is consistent with Panjiva assigning TEU through a constant linear conversion factor applied to gross weight within HS6, $\mathrm{TEU}_i \approx c_{h6(i)} \cdot \mathrm{KG}_i$. First, HS6 fixed effects alone absorb 92.5% of the shipment-level variance of the ratio $\mathrm{TEU}_i/\mathrm{kg}_i$, where $\mathrm{kg}_i$ denotes gross weight. Second, Table G.6 shows that shipment weight is proportional to TEU at a constant rate, as changes in $\log \mathrm{kg}_i$ have no economically meaningful effect on the implied conversion factor. Estimating

$$\log(\mathrm{TEU}_i/\mathrm{kg}_i) = \alpha_{h(i)} + \beta \log \mathrm{kg}_i + \varepsilon_i, \tag{28}$$

where $h(i)$ denotes the HS cell of shipment $i$, yields $\hat{\beta} = -0.0008$ at the HS6 level: statistically nonzero but economically negligible, since doubling shipment weight moves the implied conversion factor by less than 0.1 percent, and shipment size explains 0.1 percent of the within-HS6 variance of the ratio.

Table G.6: Proportionality of the Panjiva TEU–Weight Conversion Within HS Codes

| Fixed effects | HS6 |
|---|---|
| $\log$ kg ($\beta$) | -0.0008 |
| | (0.0000) |
| $\Delta$ ratio, size doubling (%) | -0.06 |
| Within $R^2$ (%) | 0.11 |
| Number of groups | 4,880 |
| Observations | 31,554,169 |

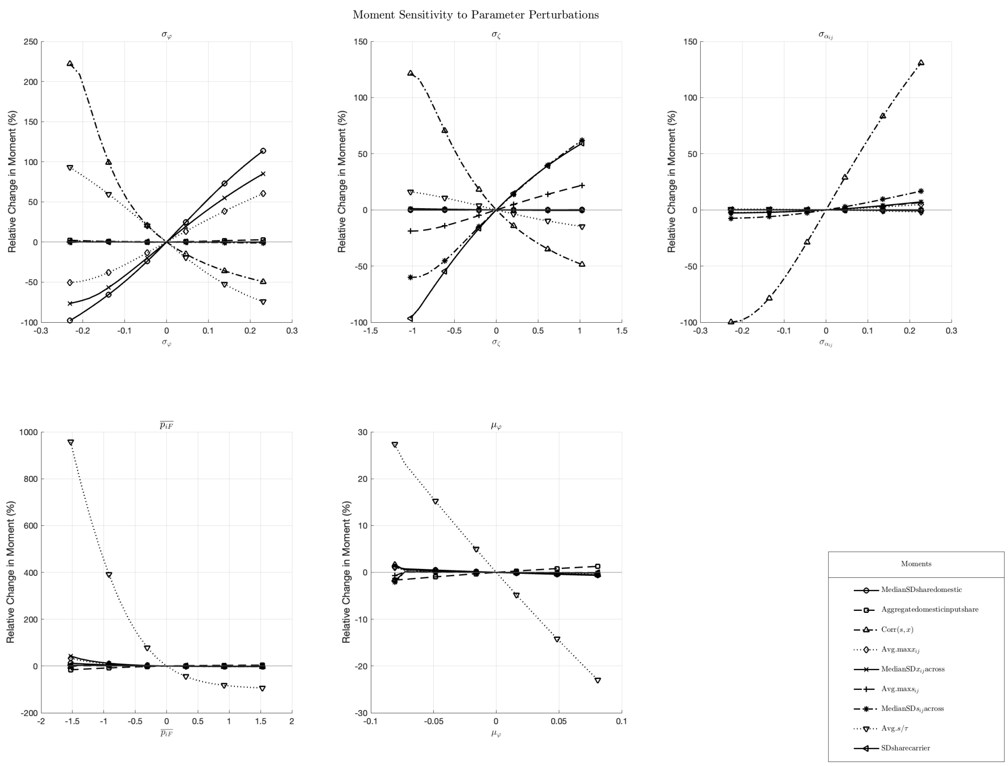
**Notes**: The Table reports estimates of $\log(\mathrm{TEU}_i/\mathrm{kg}_i) = \alpha_{h(i)} + \beta \log \mathrm{kg}_i + \varepsilon_i$, where $\mathrm{TEU}_i$ is the Panjiva `volumeteu` field, $\mathrm{kg}_i$ is shipment gross weight, and $h(i)$ denotes the HS aggregation level absorbed by fixed effects. Under an exactly proportional mapping $\mathrm{TEU}_i = c_{h(i)} \cdot \mathrm{kg}_i$ the coefficient $\beta$ is zero. The sample is the original Panjiva sample of maritime shipments, restricted to shipments with strictly positive TEU and gross weight; singleton HS groups are dropped. Standard errors are clustered at the HS level absorbed.

Therefore, to create a HS6-level mapping, we assign to each HS6 code the median ratio $\mathrm{TEU}_i/\mathrm{kg}_i$ across shipments. Results are nevertheless robust to this choice.

Lastly, because TEU is a measure that is mostly relevant for maritime shipping containers, we restrict the sample to maritime transportation and exclude HS2 codes that are usually shipped either as dry bulk or by tanker. We follow the classification provided in Arslanalp et al. (2025).

# H Additional Model Results

Figure H.1: Model identification: sensitivity of moments to parameter variation



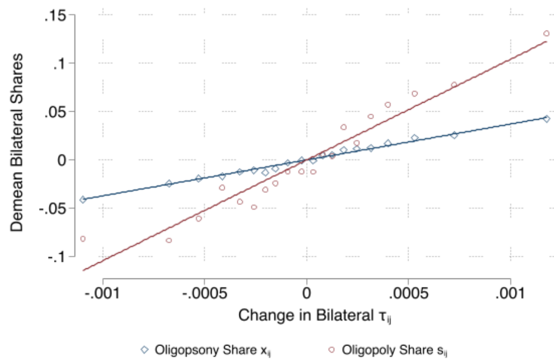

**Notes**: The figure illustrates how each simulated moment responds to variation in a single parameter, holding all other parameters fixed at their estimated values. Each moment is normalized to zero at the baseline parameter value. The horizontal axis reports the deviation of the parameter from its estimated value while the vertical axis reports the corresponding percentage change in the simulated moment.

Table H.1: Sensitivity of Moments to Parameters ($\Lambda$)

| | Moments | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Med. SD share dom. | Agg. dom. share | Corr$(s, x)$ | Avg. max $x_{ij}$ | Med. SD $x_{ij}$ | Avg. max $s_{ij}$ | Med. SD $s_{ij}$ | Avg. $s/\tau$ | SD share carrier |
| $\sigma_\varphi$ | -0.186 | -0.108 | -0.012 | 0.153 | -0.035 | 0.477 | -0.108 | -0.002 | -0.103 |
| $\sigma_\zeta$ | -0.094 | -0.050 | 0.004 | -1.786 | 1.945 | 1.513 | -0.790 | 0.004 | -1.330 |
| $\sigma_{\alpha_{ij}}$ | -0.001 | 0.001 | -0.126 | 0.051 | -0.245 | -0.673 | -0.107 | 0.007 | 0.241 |
| $\overline{p_{iF}}$ | 0.220 | 15.429 | 0.055 | -0.611 | 1.443 | -3.330 | 0.870 | 1.056 | 0.904 |
| $\mu_\varphi$ | 0.113 | -9.669 | -0.023 | 0.498 | -1.129 | 1.516 | -0.358 | -0.316 | -0.419 |

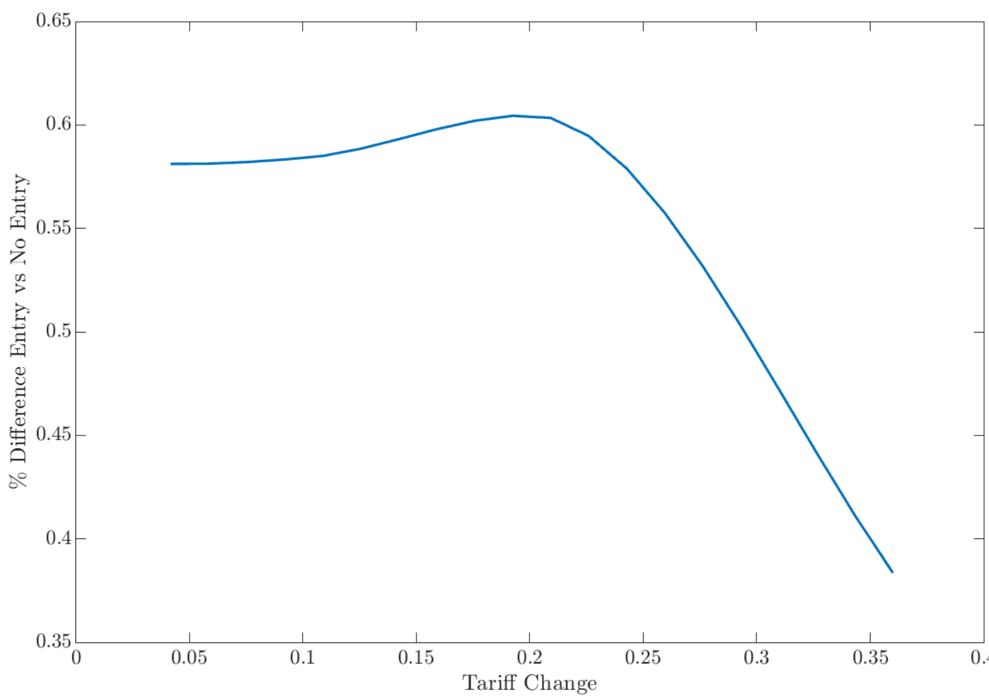
**Notes:** Each entry $\Lambda_{km}$ reports the approximate change in the estimate of parameter $k$ when moment $m$ is shifted by one unit, holding all other moments fixed, following Andrews et al. (2017). $\Lambda = -(G'WG)^{-1}G'W$, where $G$ is the numerical Jacobian of the model moments with respect to the parameters evaluated at the SMM estimates and $W$ is the optimal weighting matrix.

Figure H.2: Heterogeneity in Bilateral Prices Changes




**Notes**: The figure plots the relationship between the change in bilateral unit freight prices $\tau_{ij}$ due to the introduction of a 10 percent tariff on imports and the bilateral shares $s_{ij}$ and $x_{ij}$ in the initial equilibrium. The unit of observation is a carrier-importer pair. We absorb carrier-simulation and importer-simulation fixed effects.

Figure H.3: Extensive Margin and GFT



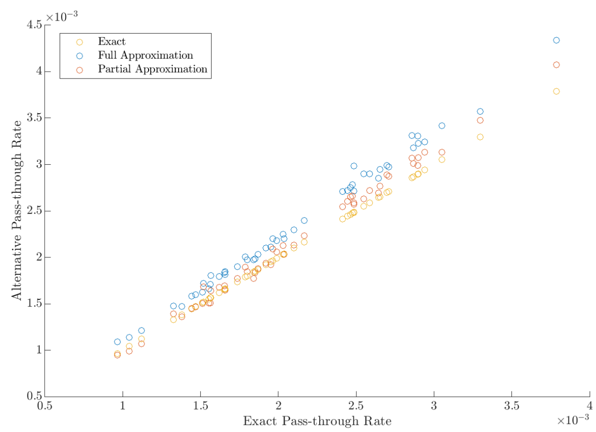

**Notes**: The figure plots the percentage difference between the change in consumers' price index accounting for entry and exit of importers relative to the change in consumers' price index without extensive margin. The percentage difference is computed for different level of tariff changes between 5 and 35 per cent. Welfare changes are computed for the baseline model, averaging across simulated economies.

Table H.2: Pass-through of Tariffs

| Model | Description | Bargaining | RTS | Aggregate PT |
|---|---|---|---|---|
| (1) | Baseline | Yes | Yes | 0.21 |
| (2) | Iceberg | No | No | 0.32 |
| (3) | Additive | No | No | 0.29 |
| (4) | $\phi = 0$ | No | Yes | 0.17 |
| (5) | $\phi = 1$ | Yes | Yes | 0.23 |
| (6) | $\theta = 1$ | Yes | No | 0.17 |
| (7) | $\theta = 1\&\phi = 0$ | No | No | 0.10 |

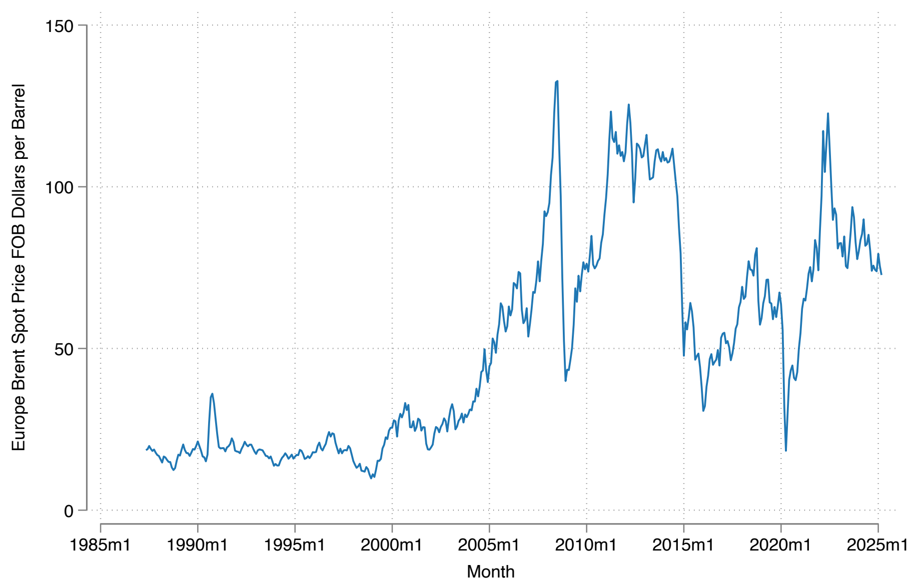
**Notes**: The table displays the percentage change in welfare due to a 10 percent tariff increase for different values of importer bargaining power, $\phi$, and carrier's return to scale, $\theta$. Welfare is measured as the consumer aggregate price index. Numbers are averaged across simulated economies.

Figure H.4: Comparison Exact vs Approximated GFT




**Notes**: The figure displays the relationship between the exact change in consumers' aggregate price index and the one computed using the approximation derived in Appendix A.4 for the baseline model. Blue circles report the approximation that used both Equations (17) and (18), while the red circles combine the vector g computed numerically into Equations (17). Each circle represents one simulated economy.

Figure H.5: Brent Index Evolution




**Notes**: Variation in the Brent Index. Source: U.S. Energy Information Administration.

During the preparation of this work the authors used ChatGPT and Claude in order to improve overall readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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